MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TIMER, DIGITAL, KEY CODE YKV

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic timer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.198 failures per million operating hours for class I or 0.723 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be YKV.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/274-1 for class I and M28787/274-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 4.5 to 18 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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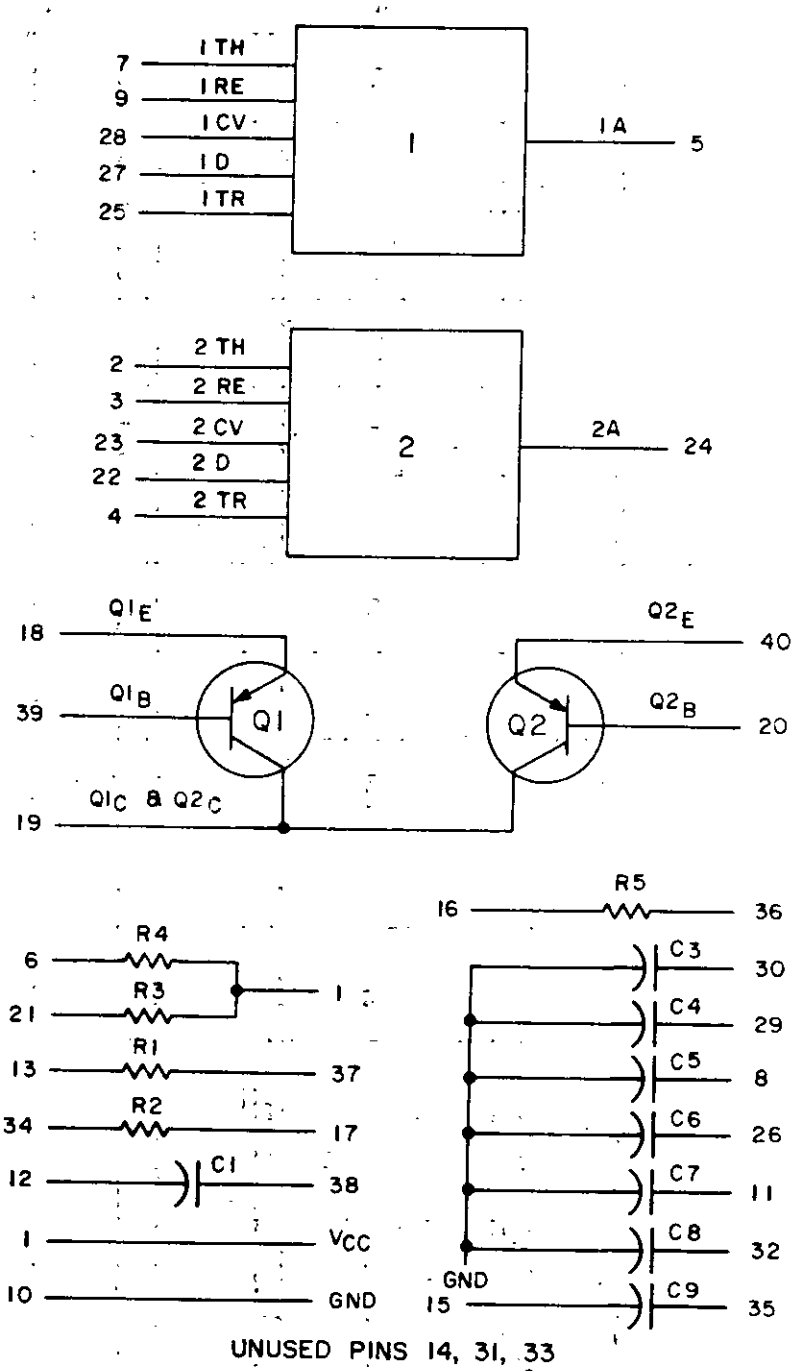


FIGURE 1. YKV module function diagram, timer.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/ Min	25°C ±5°C 4/ Max	Class I 4/ 0°C ±5°C Min	Class I 4/ 0°C ±5°C Max	Class I 4/ 60°C ±5°C Min	Class I 4/ 60°C ±5°C Max	Class II 4/ -55°C ±5°C Min	Class II 4/ -55°C ±5°C Max	Class II 4/ 85°C ±5°C Min	Class II 4/ 85°C ±5°C Max	Unit
A1, A2	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1, B2	V_{OUT} (low-loaded)	III	S	0.01	3.00	0.01	3.00	0.01	3.00	0.01	3.00	0.01	3.00	V
C1, C2	Output voltage, low level (V_{CC} = 5 V), I_{OL} = 5 mA	IV	X	0	250	0	350	0	350	0	350	0	350	mV
D1, D2	Output voltage, high level (V_{CC} = 5 V), I_{OH} = 100 mA	V	X	2.7	5.0	3.1	5.0	2.7	5.0	3.1	5.0	2.7	5.0	V
E1, E2	Output voltage, low level (V_{CC} = 15 V), I_{OL} = 10 mA	VI	X	0	150	0	150	0	250	0	150	0	250	mV
E3, E4	Output voltage, low level (V_{CC} = 15 V), I_{OL} = 50 mA	VI	X	0	500	0	500	0	700	0	500	0	700	mV
E5, E6	Output voltage, low level (V_{CC} = 15 V), I_{OL} = 100 mA	VI	X	0.0	2.2	0.0	2.8	0.0	2.8	0.0	2.8	0.0	2.8	V
F1, F2	Output voltage, high level (V_{CC} = 15 V), I_{OH} = 100 mA	VII	X	12.5	15.0	13.1	15.0	12.5	15.0	13.1	15.0	12.5	15.0	V
G1, G2	Trigger voltage check	VIII	X	4.7	5.3	4.5	5.3	4.7	5.6	4.5	5.3	4.7	5.6	V
G3, G4	Threshold voltage check	VIII	X	9.7	10.3	9.6	10.4	9.6	10.4	9.6	10.4	9.6	10.4	V
G5, G6	Reset voltage check	VIII	X	0.1	1.3	0.1	1.3	0.1	1.3	0.1	1.3	0.1	1.3	V
G7, G8	Control voltage check	VIII	X	9.8	10.2	9.8	10.2	9.8	10.2	9.8	10.2	9.8	10.2	V
H1, H2	Input current	IX	X	0.00	0.25	0.00	2.5	0.00	0.25	0.00	2.5	0.00	0.25	µA
H3, H4	Input current	IX	X	0	-5	0	-5	0	-5	0	-5	0	-5	µA
H5, H6	Input current	IX	X	-0.01	-1.60	-0.01	-1.60	-0.01	-1.60	-0.01	-1.60	-0.01	-1.60	mA
J1, J2	Saturation voltage of discharge transistor	X	X	0	800	0	800	0	1000	0	800	0	1000	mV
K1, K2	Discharge leakage current	XI	X	0.0	0.1	0.0	0.1	0.0	3.0	0.0	0.1	0.0	3.0	µA
L	$Q1_C$ and $Q2_C$ saturation voltage	XII	X	4.75	5.00	4.75	5.00	4.75	5.00	4.75	5.00	4.75	5.00	V

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M	$Q1_C$ and $Q2_C$ leakage current	XIII	X	0	200	0	200	0	200	0	200	0	200	nA
N1	Power supply current, low level (V_{CC} = 5 V)	XIV	X	1	10	1	10	1	10	1	10	1	10	mA
N2	Power supply current, low level (V_{CC} = 15 V)	XIV	X	2	30	2	30	2	30	2	30	2	30	mA
N3	Power supply current, high level (V_{CC} = 5 V)	XIV	X	1	9	1	9	1	9	1	9	1	9	mA
N4	Power supply current, high level (V_{CC} = 15 V)	XIV	X	2	21	2	21	2	21	2	21	2	21	mA
P1, P2	Initial accuracy of monostable timing (V_{CC} = 5 V)	XV	X	72.9	80.0	72.7	78.0	73.2	81.0	72.3	78.0	73.3	81.3	µs
Q1, Q2	Initial accuracy of monostable timing (V_{CC} = 15 V)	XVI	X	71.4	78.4	71.2	78.2	71.6	78.7	70.8	77.8	71.8	78.9	µs
R1, R2	Initial accuracy of astable timing (V_{CC} = 5 V)	XVII	X	19.7	24.6	19.6	24.5	19.8	24.7	19.5	24.1	19.9	24.8	µs
R3, R4	Initial accuracy of astable timing (V_{CC} = 5 V)	XVII	X	9.83	12.3	9.81	12.2	9.86	12.4	9.76	12.1	9.89	12.5	µs
S1, S2	Initial accuracy of astable timing (V_{CC} = 15 Vc)	XVIII	X	20.1	25.1	20.0	25.0	20.2	25.2	19.9	24.9	20.3	25.3	µs
S3, S4	Initial accuracy of astable timing (V_{CC} = 15 V)	XVIII	X	10.03	12.50	10.0	12.4	10.06	12.60	9.96	12.30	10.09	12.70	µs
T1, T13	Monostable output pulse width R = 10 kΩ , C = 0.01 µF	XIX	S	96.1	124.6	69.3	141.6	69.0	142.4	69.3	141.6	68.8	142.8	µs
T2, T14	Monostable output pulse width R = 49.9 kΩ, C = 0.01 µF	XIX	S	479.3	622.0	346.2	706.9	344.4	710.3	346.2	706.9	343.3	712.5	µs
T3, T15	Monostable output pulse width R = 100 kΩ, C = 0.01 µF	XIX	S	0.96	1.25	0.694	1.42	0.690	1.42	0.694	1.42	0.688	1.43	ms
T4, T16	Monostable output pulse width R = 10 kΩ, C = 0.1 µF	XIX	S	0.96	1.25	0.694	1.42	0.690	1.42	0.694	1.42	0.688	1.43	ms
T5, T17	Monostable output pulse width R = 49.9 kΩ, C = 0.1 µF	XIX	S	4.79	6.22	3.46	7.07	3.44	7.10	3.46	7.07	3.43	7.13	ms
T6, T18	Monostable output pulse width R = 100 kΩ, C = 0.1 µF	XIX	S	9.6	12.5	6.94	14.16	6.90	14.24	6.94	14.16	6.88	14.28	ms

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T7, T19	Monostable output pulse width R = 10 kΩ, C = 0.22 µF	XIX	S	2.11	2.74	1.53	3.12	1.52	3.13	1.53	3.12	1.51	3.14	ms
T8, T20	Monostable output pulse width R = 49.9 kΩ, C = 0.22 µF	XIX	S	10.5	13.7	7.62	15.55	7.57	15.63	7.62	15.55	7.55	15.67	ms
T9, T21	Monostable output pulse width R = 100 kΩ, C = 0.22 µF	XIX	S	21.1	27.4	15.26	31.16	15.2	31.3	15.26	31.16	15.1	31.4	ms
T10, T22	Monostable output pulse width R = 10 kΩ, C = 0.47 µF	XIX	S	4.51	5.86	3.26	6.66	3.24	6.70	3.26	6.66	3.23	6.72	ms
T11, T23	Monostable output pulse width R = 49.9 kΩ, C = 0.47 µF	XIX	S	22.5	29.2	16.24	33.25	16.16	33.4	16.24	33.25	16.12	33.5	ms
T12, T24	Monostable output pulse width R = 100 kΩ, C = 0.47 µF	XIX	S	45.1	58.6	32.6	66.6	32.4	67.0	32.6	66.6	32.3	67.2	ms
U1, U25	Astable output pulse characteristics, PWA	XX	S	110.5	170.2	79.9	193.4	79.4	194.4	79.9	193.4	79.2	195.0	µs
U2, U26	Astable output pulse characteristics, PWB	XX	S	55.3	85.1	39.9	96.7	39.7	97.2	39.9	96.7	39.6	97.5	µs
U3, U27	Astable output pulse characteristics, PWA	XX	S	331.3	510.0	239.2	579.4	238.0	582.3	239.2	579.4	237.3	584.1	µs
U4, U28	Astable output pulse characteristics, PWB	XX	S	275.9	424.8	199.3	482.7	198.3	485.1	199.3	482.7	197.7	486.6	µs
U5, U29	Astable output pulse characteristics, PWA	XX	S	1.11	1.70	0.799	1.93	0.794	1.94	0.799	1.93	0.792	1.95	ms
U6, U30	Astable output pulse characteristics, PWB	XX	S	0.553	0.851	0.399	0.967	0.397	0.972	0.399	0.967	0.396	0.975	ms
U7, U31	Astable output pulse characteristics, PWA	XX	S	1.11	1.70	0.799	1.93	0.794	1.94	0.799	1.93	0.792	1.95	ms
U8, U32	Astable output pulse characteristics, PWB	XX	S	0.553	0.851	0.399	0.967	0.397	0.972	0.399	0.967	0.396	0.975	ms
U9, U33	Astable output pulse characteristics, PWA	XX	S	3.31	5.10	2.39	5.80	2.38	5.83	2.39	5.80	2.37	5.85	ms
U10, U34	Astable output pulse characteristics, PWB	XX	S	2.76	4.25	2.00	4.84	1.99	4.86	2.00	4.84	1.98	4.87	ms

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U11, U35	Astable output pulse characteristics, PWA	XX	S	11.1	17.0	8.0	19.3	7.94	19.4	8.0	19.3	7.92	19.5	ms
U12, U36	Astable output pulse characteristics, PWB	XX	S	5.53	8.51	3.99	9.67	3.97	9.72	3.99	9.67	3.96	9.75	ms
U13, U37	Astable output pulse characteristics, PWA	XX	S	2.43	3.74	1.76	4.26	1.75	4.28	1.76	4.26	1.74	4.29	ms
U14, U38	Astable output pulse characteristics, PWB	XX	S	1.22	1.87	0.878	2.13	0.874	2.14	0.878	2.13	0.872	2.15	ms
U15, U39	Astable output pulse characteristics, PWA	XX	S	7.30	11.23	5.27	12.77	5.24	12.83	5.27	12.77	5.23	12.87	ms
U16, U40	Astable output pulse characteristics, PWB	XX	S	6.08	9.36	4.39	10.64	4.37	10.69	4.39	10.64	4.36	10.72	ms
U17, U41	Astable output pulse characteristics, PWA	XX	S	24.3	37.4	17.6	42.6	17.5	42.8	17.6	42.6	17.4	42.9	ms
U18, U42	Astable output pulse characteristics, PWB	XX	S	12.2	18.7	8.8	21.3	8.7	21.4	8.8	21.3	8.7	21.5	ms
U19, U43	Astable output pulse characteristics, PWA	XX	S	5.20	8.00	3.75	9.10	3.73	9.14	3.75	9.10	3.72	9.17	ms
U20, U44	Astable output pulse characteristics, PWB	XX	S	2.60	4.00	1.87	4.55	1.86	4.57	1.87	4.55	1.85	4.58	ms
U21, U45	Astable output pulse characteristics, PWA	XX	S	15.5	23.9	11.2	27.3	11.2	27.4	11.2	27.3	11.1	27.5	ms
U22, U46	Astable output pulse characteristics, PWB	XX	S	12.9	20.0	9.35	22.7	9.3	22.8	9.35	22.7	9.27	22.9	ms
U23, U47	Astable output pulse characteristics, PWA	XX	S	52.0	80.0	37.5	91.0	37.3	91.4	37.5	91.0	37.2	91.7	ms
U24, U48	Astable output pulse characteristics, PWB	XX	S	26.0	40.0	18.7	45.5	18.6	45.7	18.7	45.5	18.5	45.8	ms
V1, V2	Maximum frequency and minimum trigger pulse check	XXI	S	12.9	13.7	12.9	13.7	12.9	13.7	12.9	13.7	12.9	13.7	µs
W1, W2	Transition time (t_{TLH}), C_L = 50 pF	XXII	S	10	300	10	300	10	300	10	300	10	300	ns
X1, X2	Transition time (t_{THL}), C_L = 50 pF	XXIII	S	10	300	10	300	10	300	10	300	10	300	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1, Y2	Propagation delay time (t_{PLH}), C_L = 50 pF (trigger-to-output)	XXIV	S	25	800	25	800	25	900	25	800	25	900	ns
Z1, Z2	Propagation delay time (t_{PHL}), C_L = 50 pF (reset-to-output)	XXV	S	0.2	1.5	0.2	1.5	0.2	2.0	0.2	1.5	0.2	2.0	µs
AA1, AA2	Output crosstalk, low level	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
BB1, BB2	Output crosstalk, high level	XXVII	Q	0	400	0	400	0	400	0	400	0	400	mV
CC	Power supply transient amplitude	XXVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
DD	Power supply filter capacitance	XXIX	Q	0.7	1.3	0.7	1.3	0.7	1.3	0.7	1.3	0.7	1.3	µF
EE1,EE3-EE6,EE8-EE10	Input capacitance	XXX	Q	2	15	2	15	2	15	2	15	2	15	pF
EE2, EE7	Input capacitance	XXX	Q	2	25	2	25	2	25	2	25	2	25	pF
FF1	C1 Capacitance	XXXI	X	243	297	175	338	175	338	175	338	175	338	pF
FF2	C3 Capacitance	XXXI	X	0.090	0.110	0.065	0.125	0.065	0.125	0.065	0.125	0.065	0.125	µF
FF3	C4 Capacitance	XXXI	X	0.198	0.242	0.143	0.275	0.143	0.275	0.143	0.275	0.143	0.275	µF
FF4	C5 Capacitance	XXXI	X	0.0090	0.0110	0.0065	0.0125	0.0065	0.0125	0.0065	0.0125	0.0065	0.0125	µF
FF5	C6 Capacitance	XXXI	X	0.0090	0.0110	0.0065	0.0125	0.0065	0.0125	0.0065	0.0125	0.0065	0.0125	µF
FF6	C7 Capacitance	XXXI	X	90	110	65	125	65	125	65	125	65	125	pF
FF7	C8 Capacitance	XXXI	X	90	110	65	125	65	125	65	125	65	125	pF
FF8	C9 Capacitance	XXXI	X	0.423	0.517	0.305	0.588	0.305	0.588	0.305	0.588	0.305	0.588	µF
GG1	R1 Resistance	XXXII	X	98.5	101.5	98.5	101.5	98.0	102.0	98.5	101.5	97.7	102.3	kΩ
GG2	R2 Resistance	XXXII	X	9.85	10.15	9.85	10.15	9.80	10.20	9.85	10.15	9.77	10.23	kΩ
GG3	R3 Resistance	XXXII	X	9.85	10.15	9.85	10.15	9.80	10.20	9.85	10.15	9.77	10.23	kΩ

See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
GG4	R4 Resistance	XXXII	X	98.5	101.5	98.5	101.5	98.0	102.0	98.5	101.5	97.7	102.3	kΩ
GG5	R5 Resistance	XXXII	X	49.15	50.65	49.15	50.65	48.90	50.90	49.15	50.65	48.74	51.06	kΩ

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A,B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through GG in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through GG shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition DD, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through DD, FF, and GG shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through GG shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1,200 maximum	1,200 maximum	mW
Output drive current:			
Low level (V_{CC} = 15 V)			
V_{OL} = 0.25 V	10.0 maximum	10.0 maximum	mA
V_{OL} = 0.7 V	50.0 maximum	50.0 maximum	mA
V_{OL} = 2.8 V	100.0 maximum	100.0 maximum	mA
(V_{CC} = 5 V)			
V_{OL} = 0.35 V	5.0 maximum	5.0 maximum	mA
High level (V_{CC} = 15 V)			
V_{OH} = 13.1 V	100 maximum	100 maximum	mA
(V_{CC} = 5 V)			
V_{OH} = 3.1 V	100 maximum	100 maximum	mA
Frequency (input trigger pulse)	75 maximum	75 maximum	kHz
Trigger pulse width	200	200	ns

* 6.5 Module function. This module contains two independent timers. Both timers are capable of producing accurate time delays or oscillation by use of external resistors and capacitors on the module which can be used with the timing circuits. Also, there are two external transistors on the module which should be used in conjunction with the timing circuits only.

6.5.1 Guaranteed accuracy of the module. When the module is operating in the monostable mode, the minimum pulse width guaranteed to the accuracy stated in 6.5 is 75 µs. Also, the module will operate in the monostable mode with a maximum trigger pulse of 75 kHz but, the accuracy of the output pulse will not be guaranteed. When the module is operating in the astable mode, the maximum guaranteed frequency to the accuracy stated in 6.5 is 30 kHz. The module will operate at frequencies above 30 kHz but the accuracy is not guaranteed.

6.5.2 Input to output relationship. The device triggers on the negative edge of a negative going pulse. The trigger pulse must be of shorter duration than the "RC" time interval; if the trigger is held low, the output will stay high until the trigger is driven high again. The reset acts as an inhibit. When the reset is above 1 volt, the device is free to function; if the reset is taken below 0.4 volts, the output is forced low. When triggered, the timer starts its cycling by driving the output pin high; simultaneously, the timing capacitor starts charging. When the capacitor voltage reaches two-thirds V_{CC}, the capacitor is discharged to ground. This drives the output low, ending the timing cycle. Also, the trigger pulse must drop below one-third V_{CC} before the timer triggers. A latch-up can be caused by a negative voltage on the output.

6.5.3 Parallel outputs. Paralleling of outputs is not permissible.

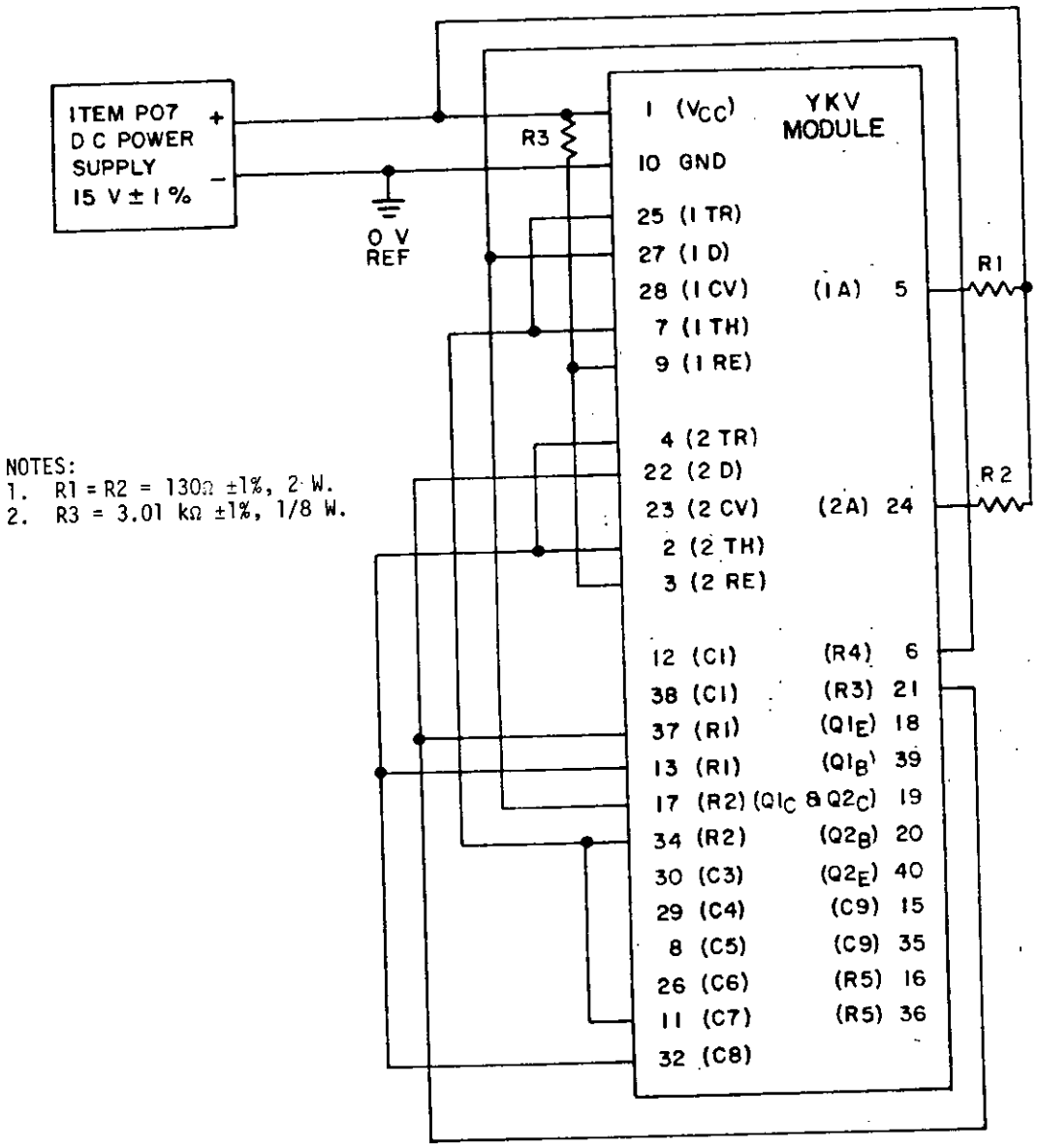
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, and deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
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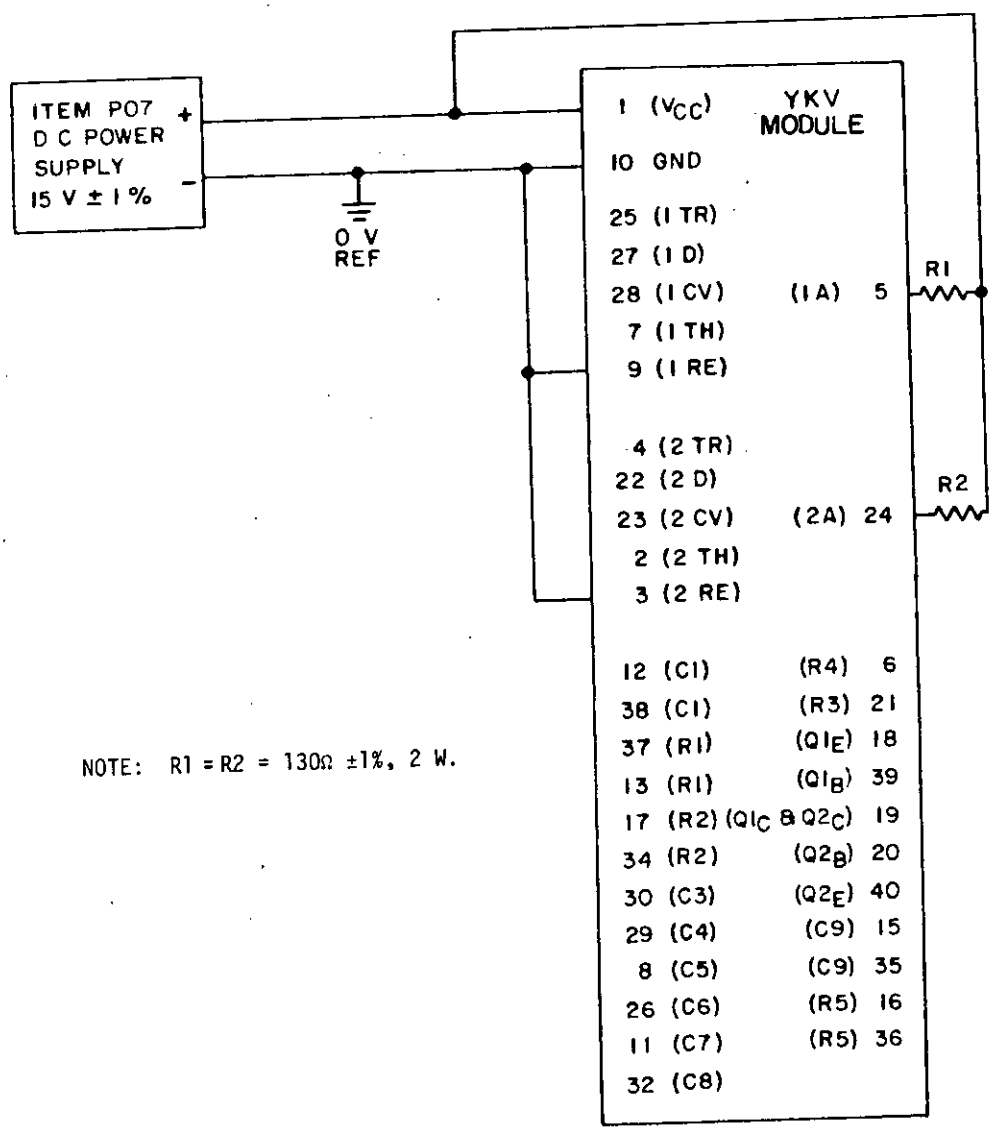
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NOTES:
1. R1 = R2 = 130Ω ±1%, 2 W.
2. R3 = 3.01 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1 = R2 = 130Ω ±1%, 2 W.

FIGURE 3. Component temperature test circuit.

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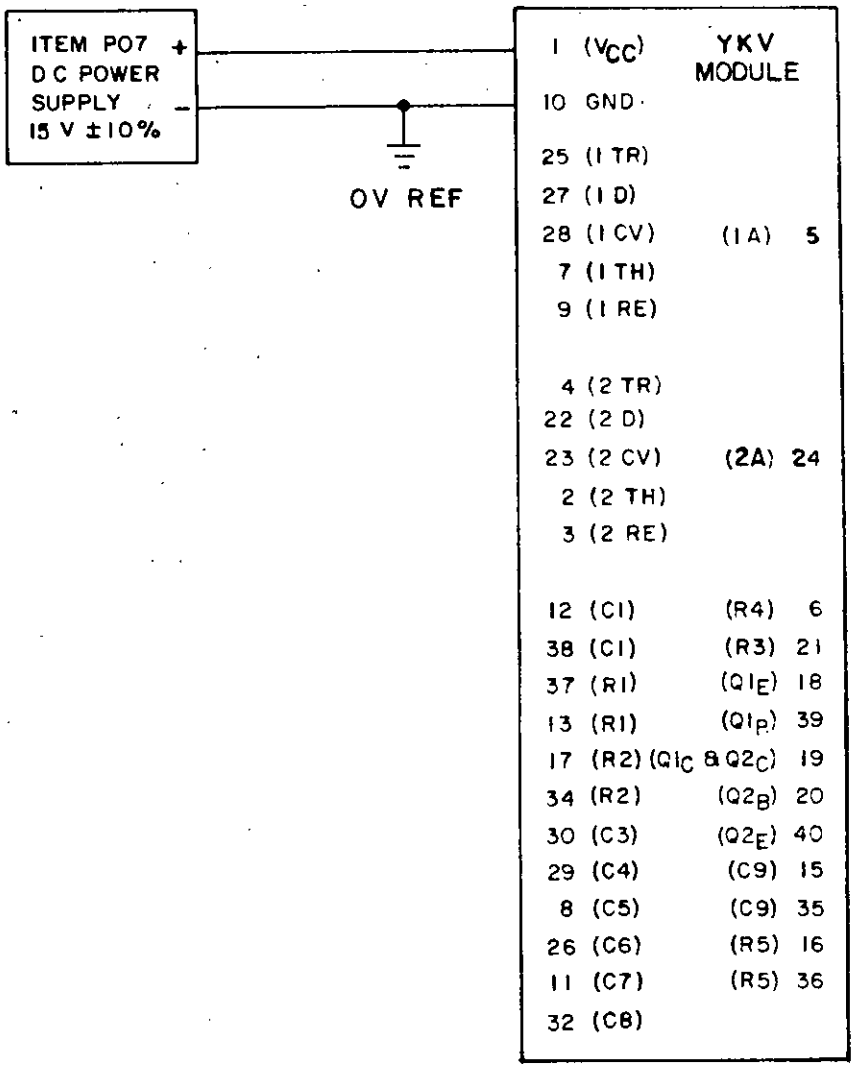


FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,D	Module frame	10	∞	MΩ	5
A2	Isolation	C	D	10	∞	MΩ	5

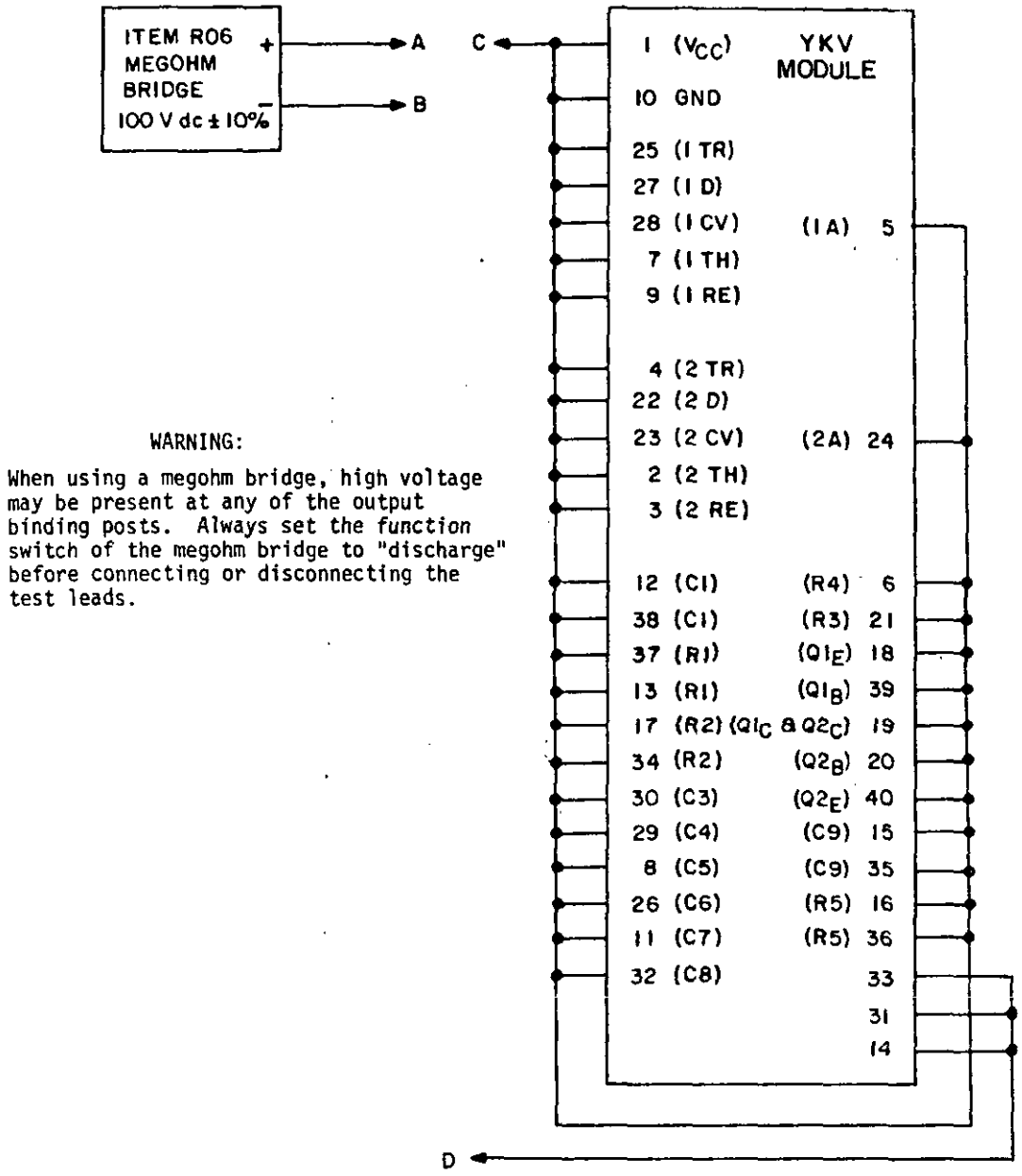
TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	OUT 1 V_{OUT} (low loaded)	5	0.01	3.00	V	6
B2	OUT 2 V_{OUT} (low loaded)	24	0.01	3.00	V	6

* TABLE IV. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	OUT 1 Output voltage, low level (V_{CC} = 5 V)	7, 25	Open	5	0	250	mV	7
C2	OUT 2 Output voltage, low level (V_{CC} = 5 V)	2, 4	Open	24	0	250	mV	7

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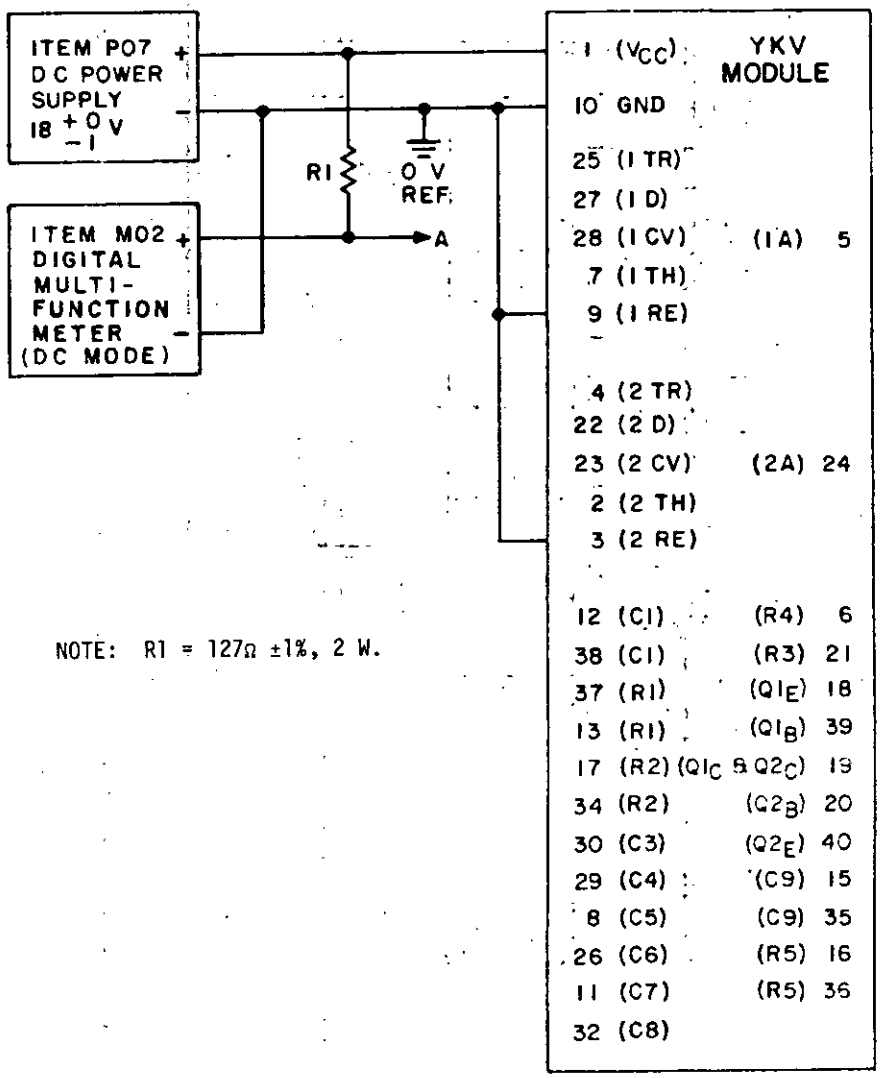
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. Pins 14, 31, and 33 are unused pins.
2. For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table II.

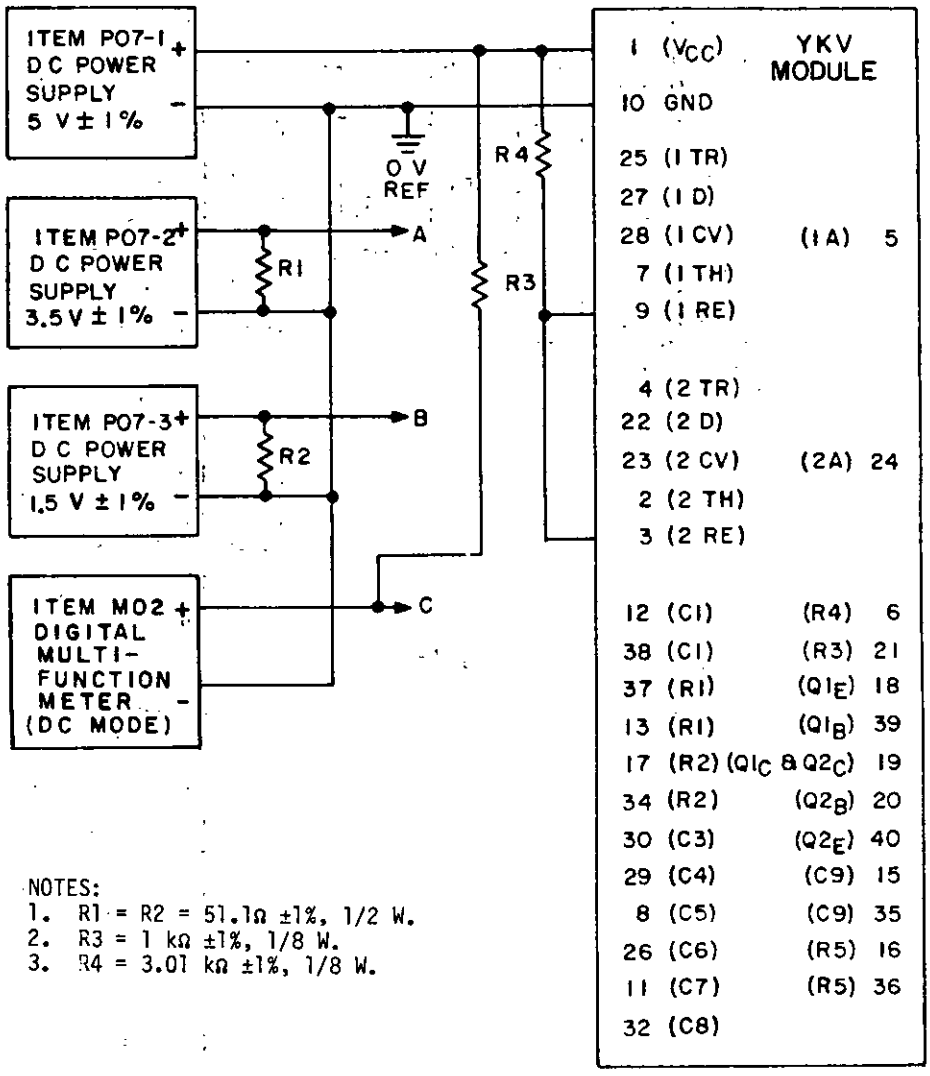
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NOTE: R1 = 127Ω ±1%, 2 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/2 W.
2. R3 = 1 kΩ ±1%, 1/8 W.
3. R4 = 3.01 kΩ ±1%, 1/8 W.

* FIGURE 7. Output voltage (low-level) test circuit for table IV.

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* TABLE V. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	OUT 1 Output voltage, high level (V_{CC} = 5 V)	7	25	5	2.7	5.0	V	8
D2	OUT 2 Output voltage, high level (V_{CC} = 5 V)	2	4	24	2.7	5.0	V	8

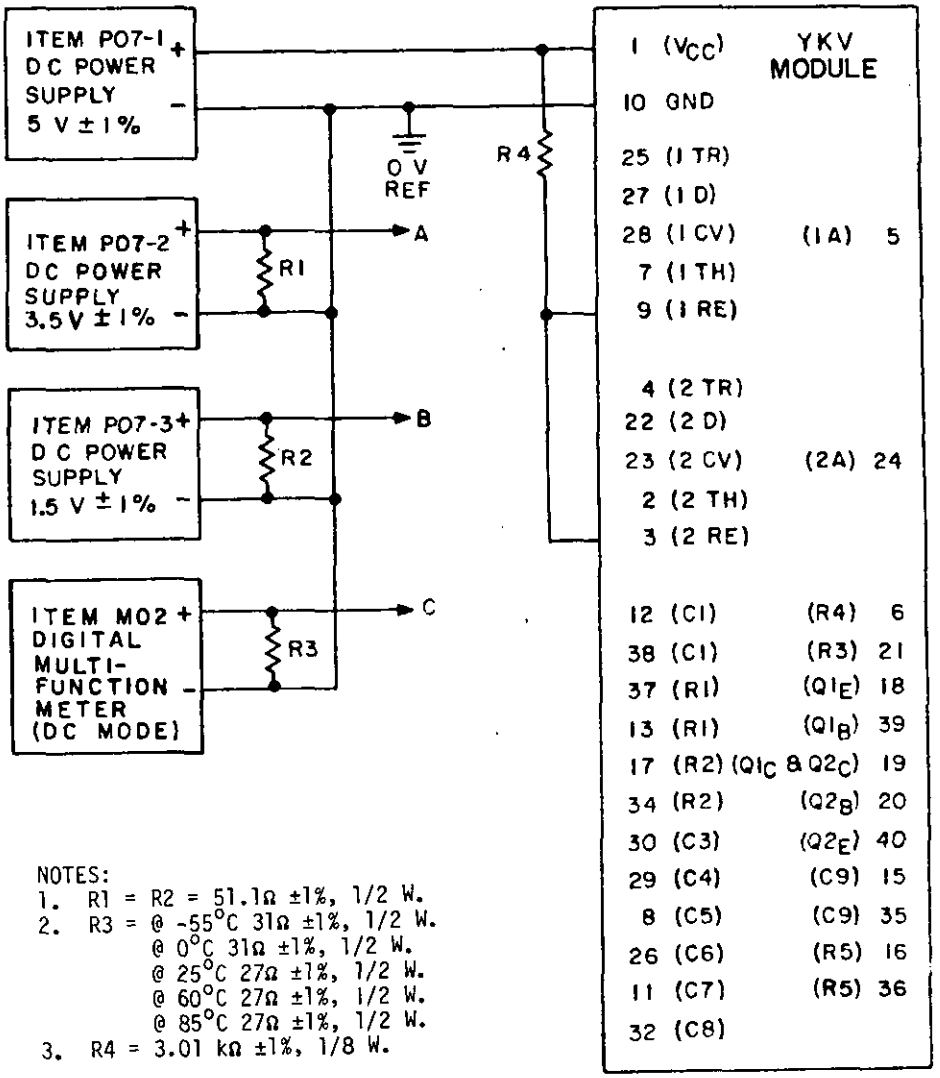
* TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	OUT 1 Output voltage, low level (V_{CC} = 15 V)	2, 7	5	0	150	mV	9
E2	OUT 2 "	"	24	"	150	"	"
E3	OUT 1 "	"	5	"	500	"	"
E4	OUT 2 "	"	24	"	500	"	"
E5	OUT 1 "	"	5	0.0	2.2	V	"
E6	OUT 2 "	"	24	0.0	2.2	V	"

* TABLE VII. Output voltage (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	OUT 1 Output voltage, high level (V_{CC} = 15 V)	4, 25	5	12.5	15.0	V	10
F2	OUT 2 Output voltage, high level (V_{CC} = 15 V)	4, 25	24	12.5	15.0	V	10

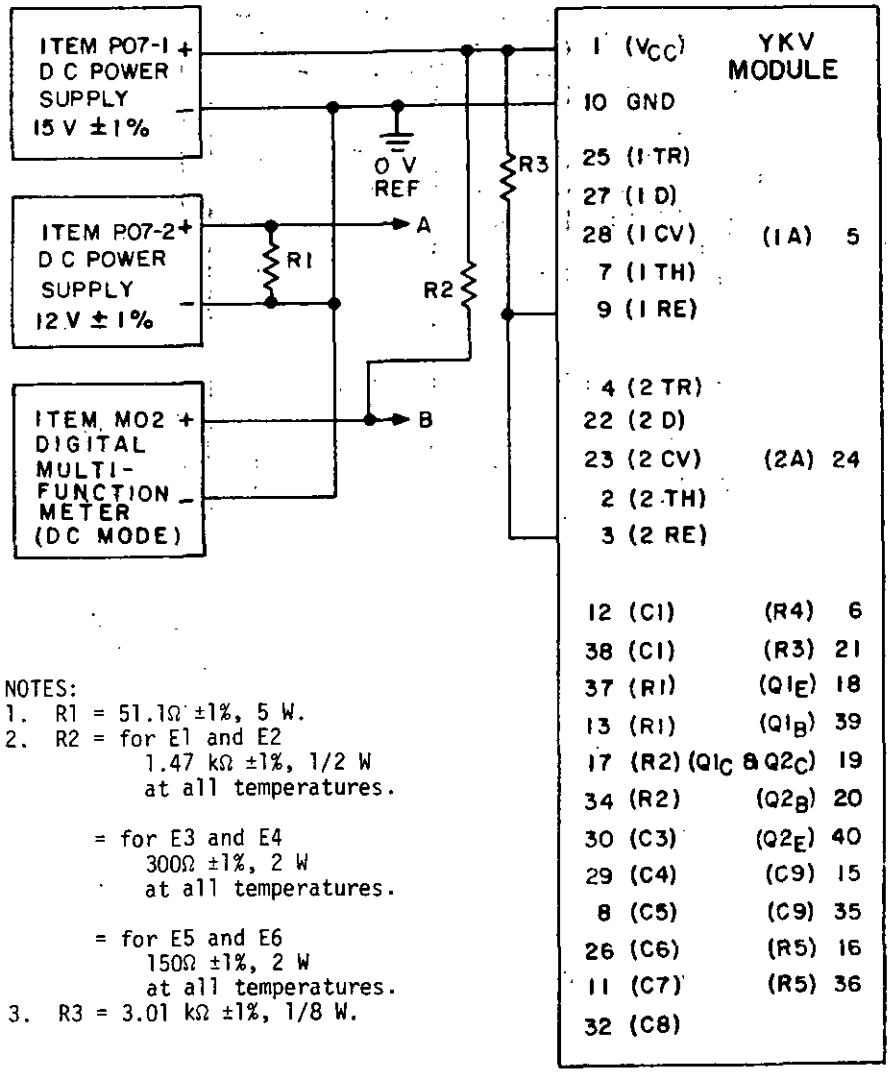
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/2 W.
2. R3 = @ -55°C 31Ω ±1%, 1/2 W.
   @ 0°C 31Ω ±1%, 1/2 W.
   @ 25°C 27Ω ±1%, 1/2 W.
   @ 60°C 27Ω ±1%, 1/2 W.
   @ 85°C 27Ω ±1%, 1/2 W.
3. R4 = 3.01 kΩ ±1%, 1/8 W.

* FIGURE 8. Output voltage (high-level) test circuit for table V.

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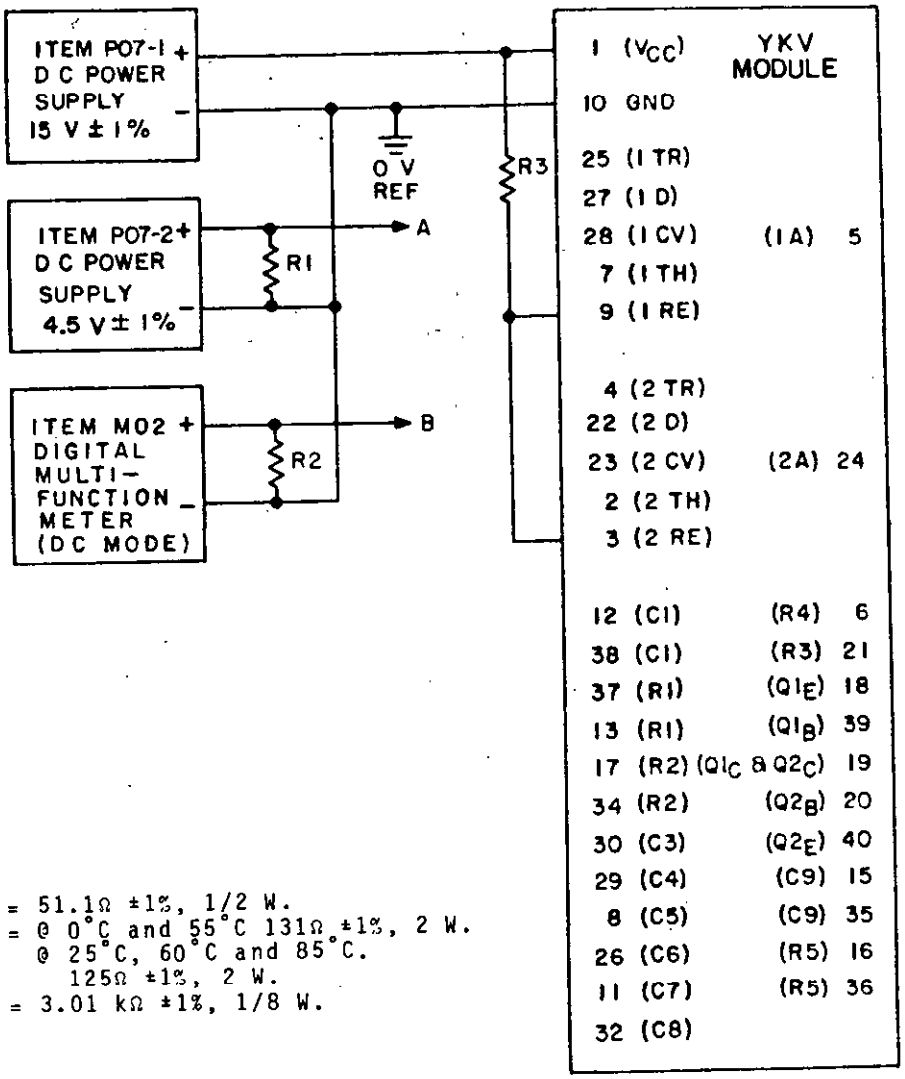
NOTES:
1. R1 = 51.1Ω ±1%, 5 W.
2. R2 = for E1 and E2
   1.47 kΩ ±1%, 1/2 W
   at all temperatures.

   = for E3 and E4
   300Ω ±1%, 2 W
   at all temperatures.

   = for E5 and E6
   150Ω ±1%, 2 W
   at all temperatures.
3. R3 = 3.01 kΩ ±1%, 1/8 W.

* FIGURE 9. Output voltage (low-level) test circuit for table VI.

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NOTES:
1. R1 = 51.1Ω ±1%, 1/2 W.
2. R2 = @ 0°C and 55°C 131Ω ±1%, 2 W.
   @ 25°C, 60°C and 85°C.
   125Ω ±1%, 2 W.
3. R3 = 3.01 kΩ ±1%, 1/8 W.

* FIGURE 10. Output voltage (high-level) test circuit for table VII.

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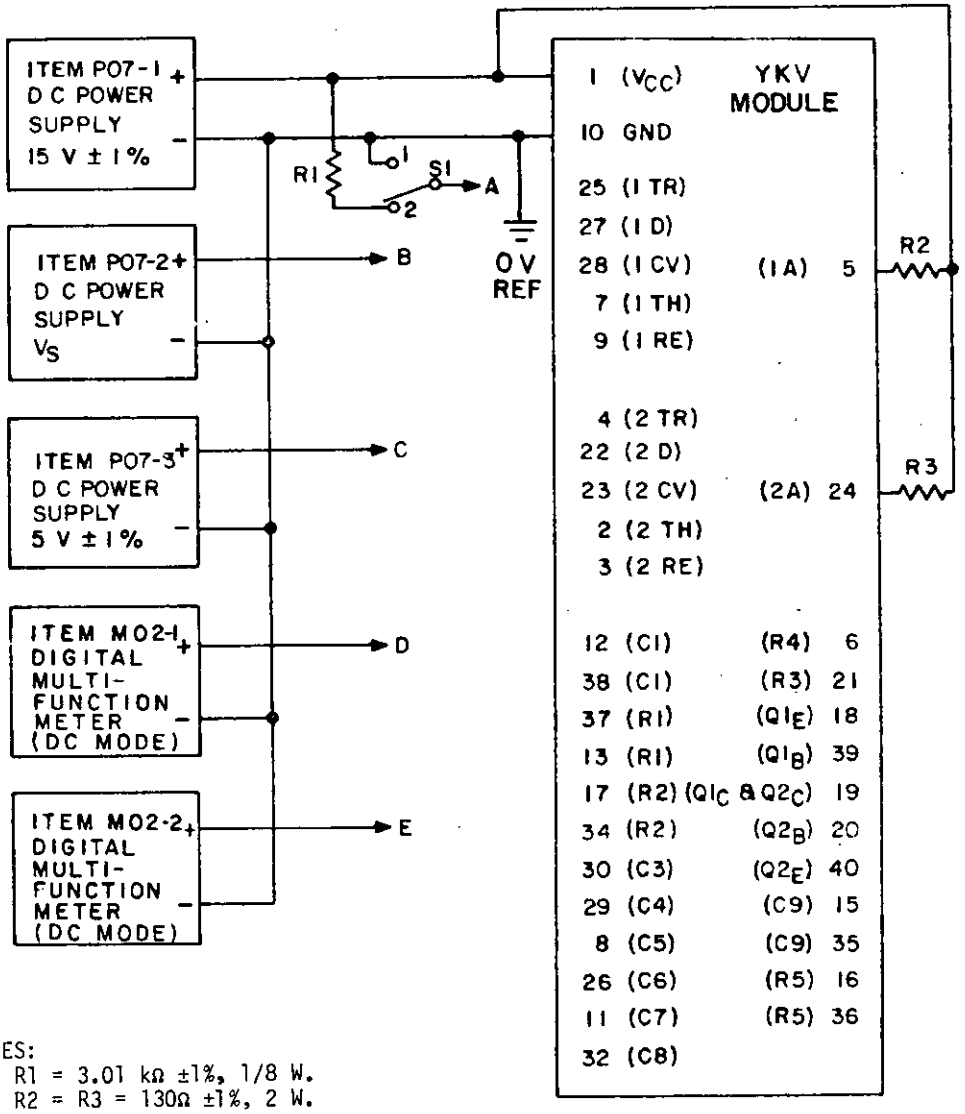
TABLE VIII. Trigger, threshold, reset, and control voltage check test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	1 TR Trigger voltage check	9, 3	25	Open	25	5	4.7	5.3	V	11
G2	2 TR Trigger voltage check	"	4	"	4	24	4.7	5.3	"	"
G3	1 TH Threshold voltage check	"	7	"	7	5	9.7	10.3	"	"
G4	2 TH Threshold voltage check	"	2	"	2	24	9.7	10.3	"	"
G5	1 RE Reset voltage check	Open	9	25	9	5	0.1	1.3	"	"
G6	2 RE Reset voltage check	"	3	4	3	24	0.1	1.3	"	"
G7	1 CV Control voltage check	"	Open	Open	28	Open	9.8	10.2	"	"
G8	2 CV Control voltage check	"	Open	Open	23	Open	9.8	10.2	"	"

TABLE IX. Input current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1 TH Input current	7	2, 25, 4, 9, 3	Open	0.00	0.25	µA	12
H2	2 TH "	2	7, 25, 4, 9, 3	"	0.00	0.25	"	"
H3	1 TR "	25	2, 7, 4, 9, 3	"	0	-5	"	"
H4	2 TR "	4	2, 7, 25, 9, 3	"	0	-5	"	"
H5	1 RE "	2, 7, 25, 4, 3	Open	9	-0.01	-1.60	mA	"
H6	2 RE "	2, 7, 25, 4, 9	Open	3	-0.01	-1.60	mA	"

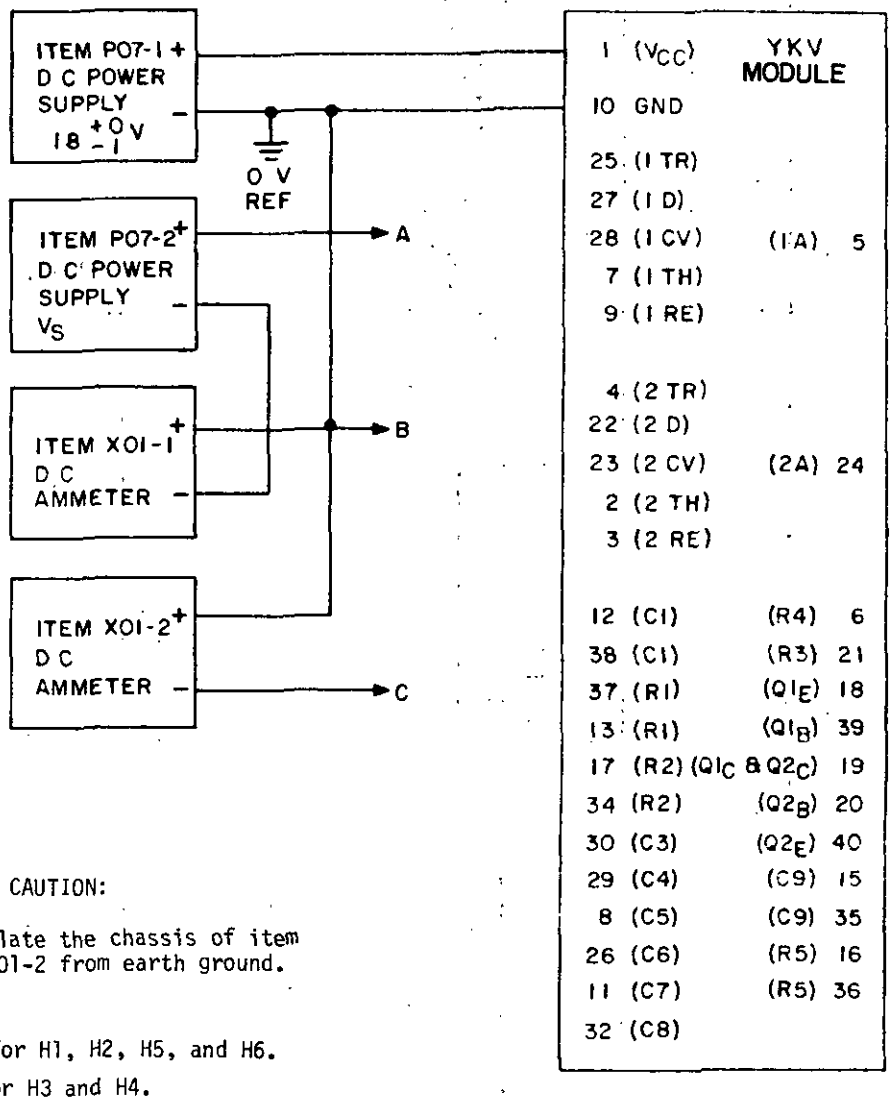
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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. R2 = R3 = 130Ω ±1%, 2 W.
3. S1 = SPDT switch.
4. For test condition G1 and G2, V_S is lowered from a voltage greater than 10 V dc. When the outputs switch to the high level state, V_{TR} is measured. Before starting either G1 or G2, place switch S1 in position 1 for a minimum of 2 µs, then switch S1 to position 2 and start test.
5. Before starting test condition G3 or G4, place pins 25 and 4 to ground for a minimum of 2 µs and then start test. For test conditions G3 and G4, V_S is increased from a voltage less than 5 V dc. When the outputs switch to the low level state, V_{TH} is measured. Place switch S1 in position 2 for these tests.
6. For test conditions G5 and G6, after placing S1 in position 2, V_S is lowered from a voltage greater than 2 V dc. When the outputs switch to the low level state, V_{RE} is then measured.

* FIGURE 11. Trigger, threshold, reset, and control voltage check test circuit for table VIII.

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CAUTION:

Do not isolate the chassis of item X01-1 or X01-2 from earth ground.

NOTES:
1. V_S = 14 V ±1% for H1, H2, H5, and H6.
2. V_S = 5 V ±1% for H3 and H4.

FIGURE 12. Input current test circuit for table IX.

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* TABLE X. Saturation voltage of discharge transistor test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
J1	1D Saturation voltage of discharge transistor	27	0	800	mV	13
J2	2D Saturation voltage of discharge transistor	22	0	800	mV	13

TABLE XI. Discharge leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1D Discharge leakage current	27	0.0	0.1	µA	14
K2	2D Discharge leakage current	22	0.0	0.1	µA	14

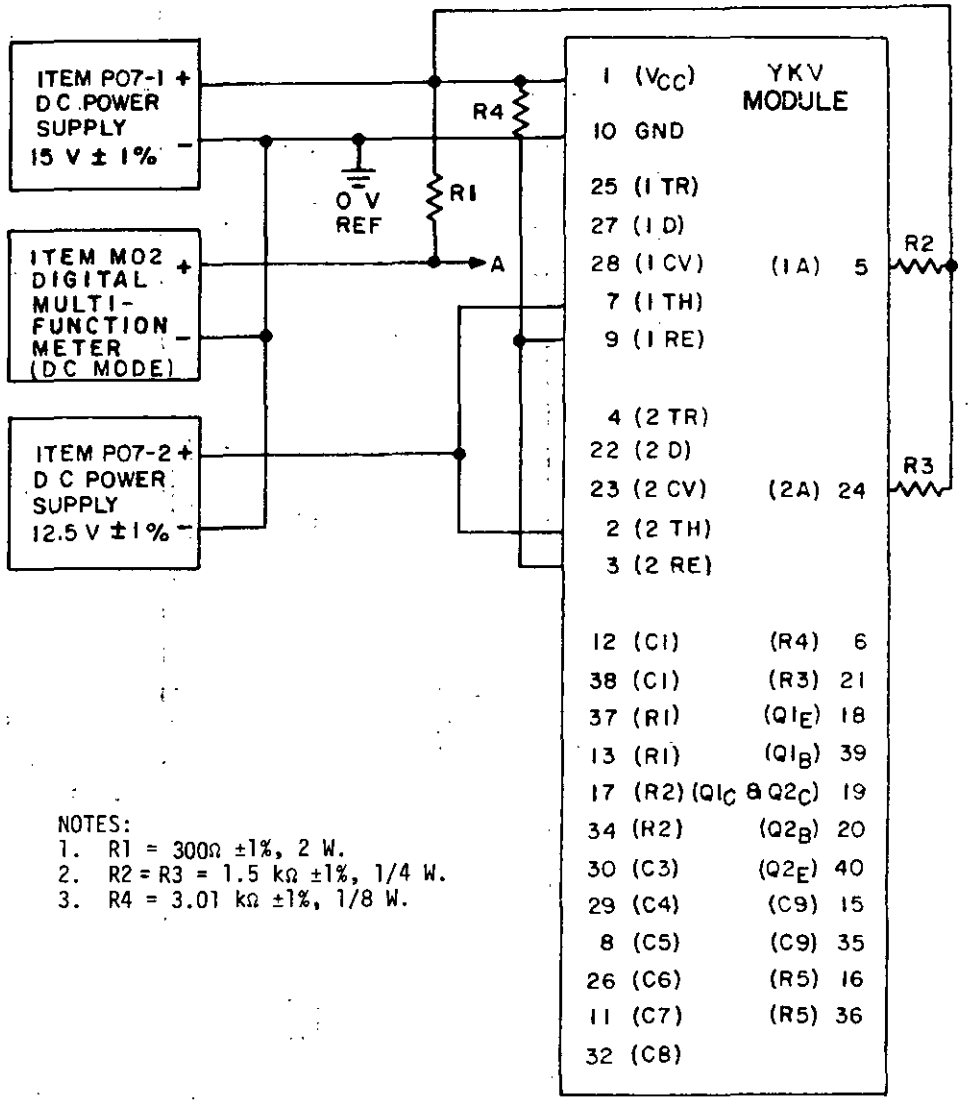
TABLE XII. $Q1_C$ and $Q2_C$ saturation voltage test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
L	$Q1_C$ and $Q2_C$ saturation voltage	4.75	5.00	V	15

TABLE XIII. $Q1_C$ and $Q2_C$ leakage current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M	$Q1_C$ and $Q2_C$ leakage current	0	200	nA	16

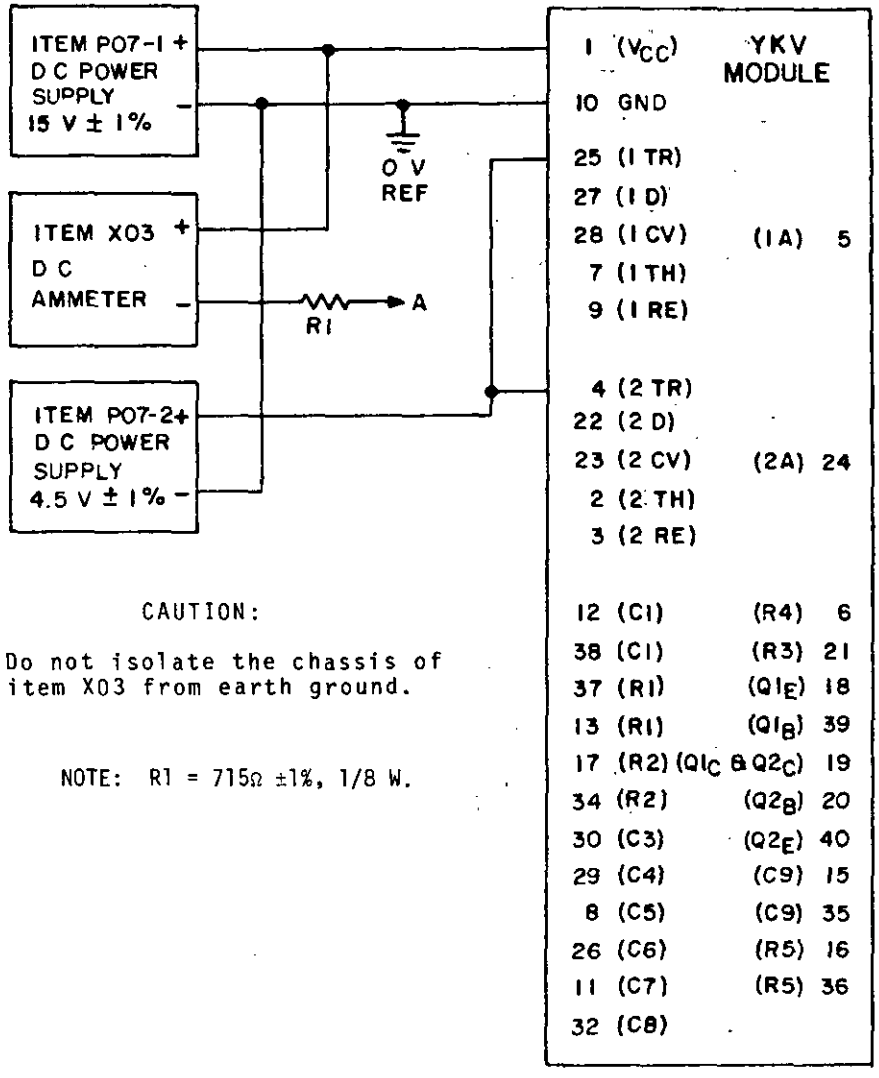
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NOTES:
1. R1 = 300Ω ±1%, 2 W.
2. R2 = R3 = 1.5 kΩ ±1%, 1/4 W.
3. R4 = 3.01 kΩ ±1%, 1/8 W.

* FIGURE 13. Saturation voltage of discharge transistor test circuit for table X.

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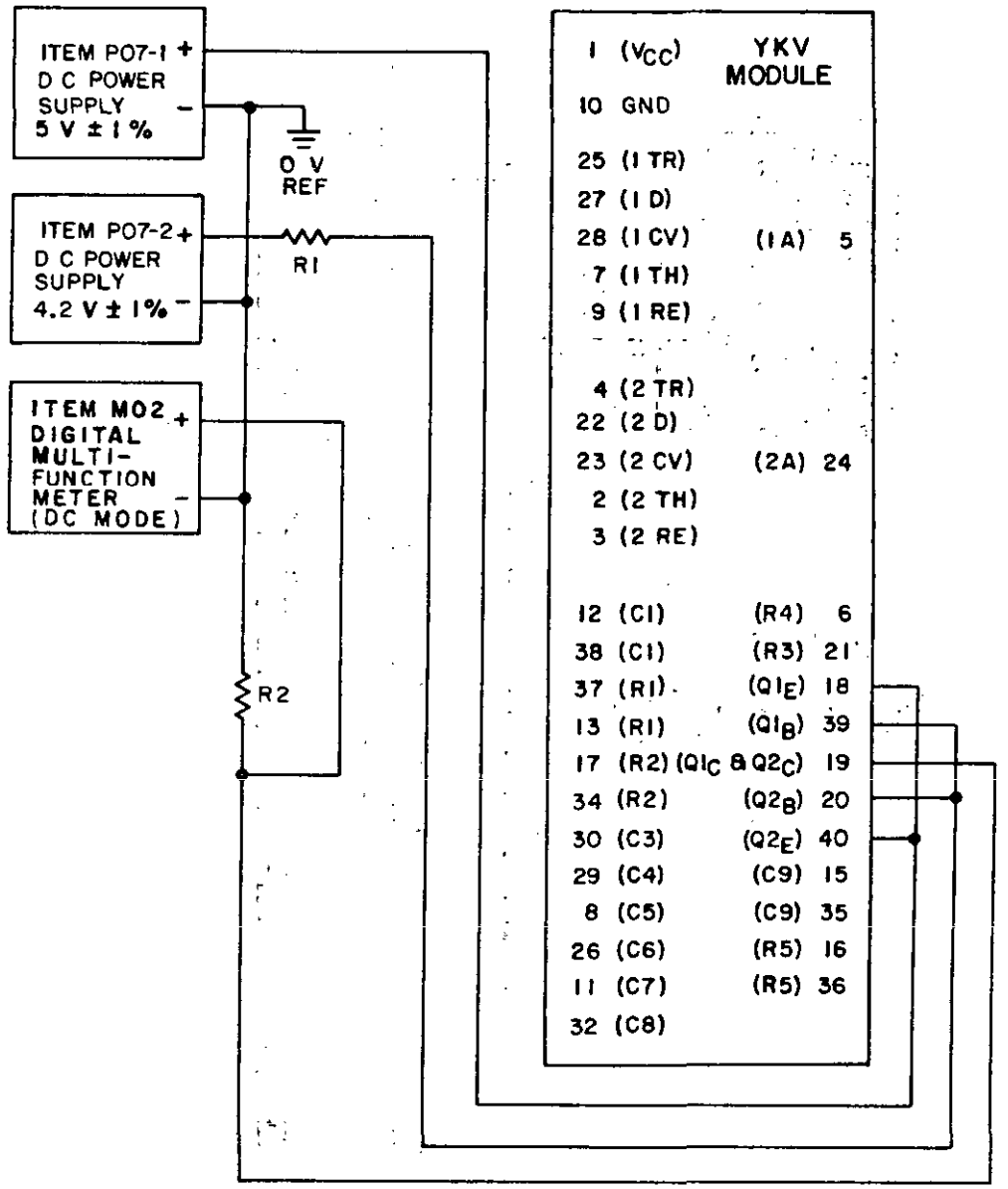
CAUTION:

Do not isolate the chassis of item X03 from earth ground.

NOTE: R1 = 715Ω ±1%, 1/8 W.

FIGURE 14. Discharge leakage current test circuit for table XI.

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NOTES:
1. R1 = 422Ω ±1%, 1/8 W.
2. R2 = 200Ω ±1%, 1/2 W.

*FIGURE 15. $Q1_C$ and $Q2_C$ saturation voltage test circuit for table XII.

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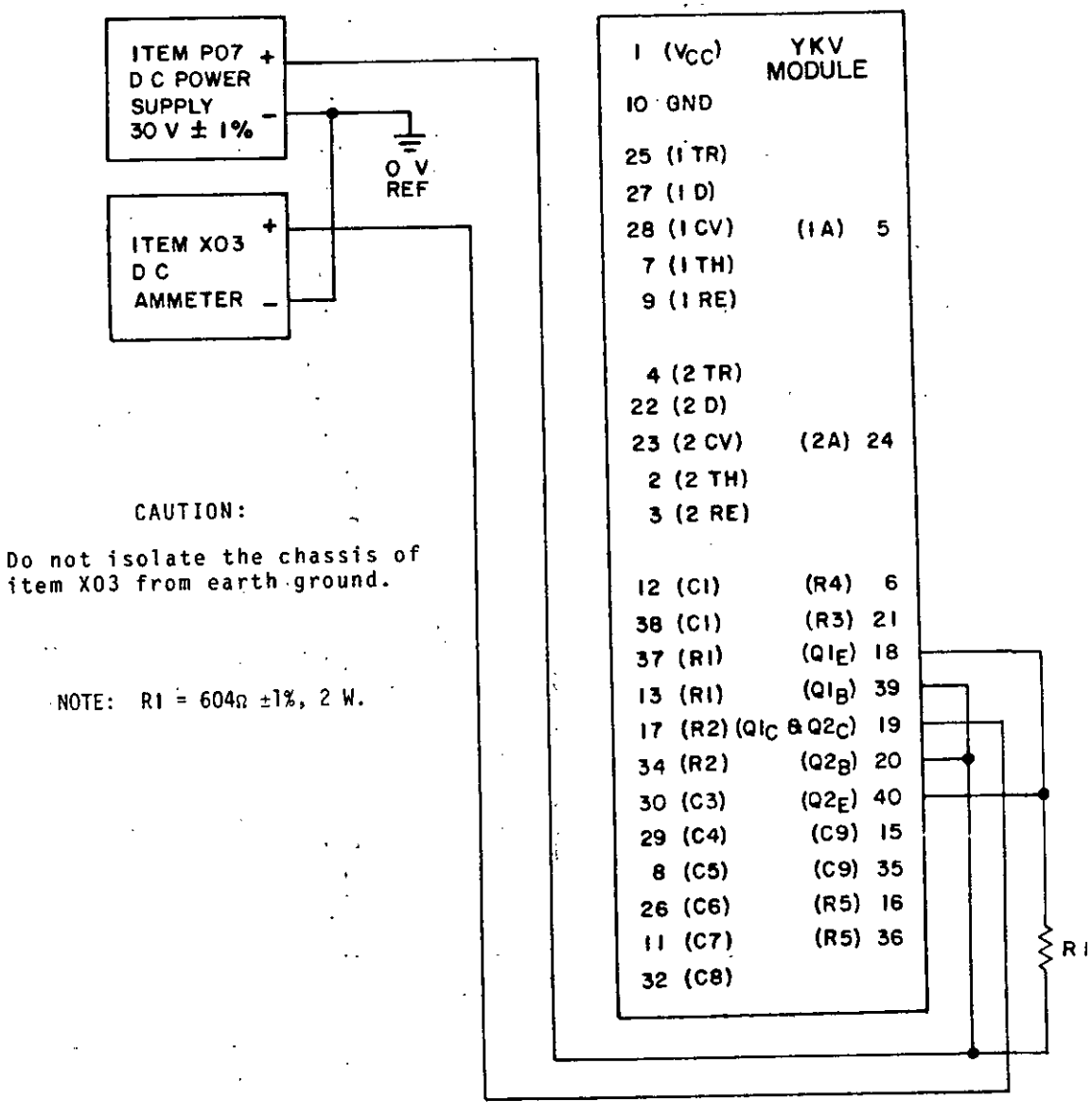


FIGURE 16. $Q1_C$ and $Q2_C$ leakage current test circuit for table XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N1	Power supply current (V_{CC} = 5 V), low level	1	10	mA	17
N2	Power supply current (V_{CC} = 15 V), low level	2	30	"	"
N3	Power supply current (V_{CC} = 5 V), high level	1	9	"	"
N4	Power supply current (V_{CC} = 15 V), high level	2	21	"	"

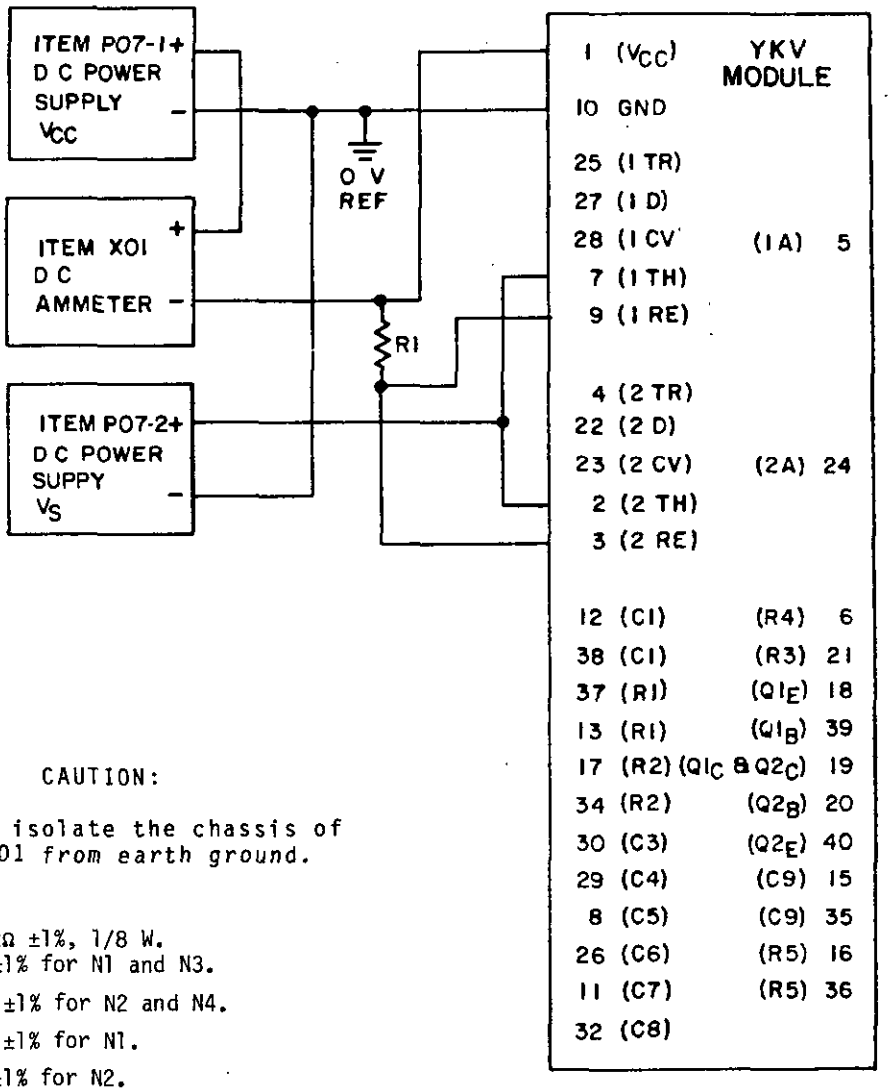
* TABLE XV. Initial accuracy of monostable timing test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	OUT 1 (PWB) Initial accuracy of monostable timing, V_{CC} = 5 V	25	5	72.9	80.0	µs	18, 19
P2	OUT 2 (PWB) Initial accuracy of monostable timing, V_{CC} = 5 V	4	24	72.9	80.0	µs	18, 19

TABLE XVI. Initial accuracy of monostable timing test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	OUT 1 (PWB) Initial accuracy of monostable timing, V_{CC} = 15 V	25	5	71.4	78.4	µs	18, 19
Q2	OUT 2 (PWB) Initial accuracy of monostable timing, V_{CC} = 15 V	4	24	71.4	78.4	µs	18, 19

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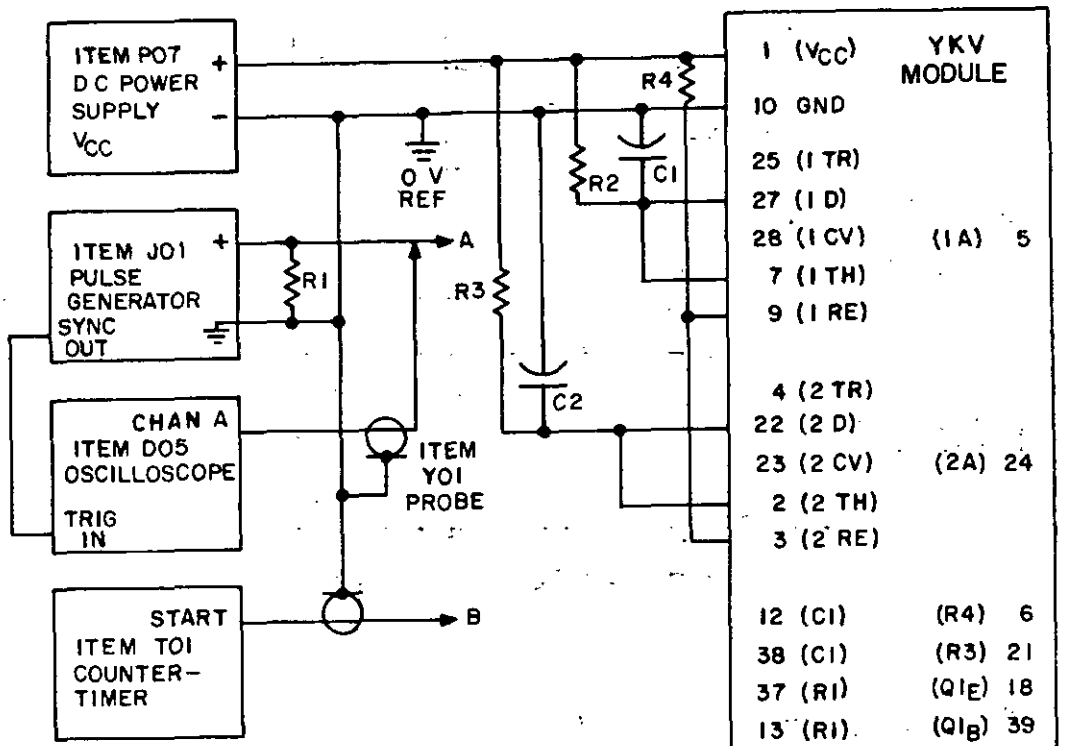
CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. V_{CC} = 5 V ±1% for N1 and N3.
   V_{CC} = 15 V ±1% for N2 and N4.
3. V_S = 3.5 V ±1% for N1.
   V_S = 12 V ±1% for N2.
   V_S = 1.5 V ±1% for N3.
   V_S = 4 V ±1% for N4.

FIGURE 17. Power supply current (low-level and high-level) test circuit for table XIV.

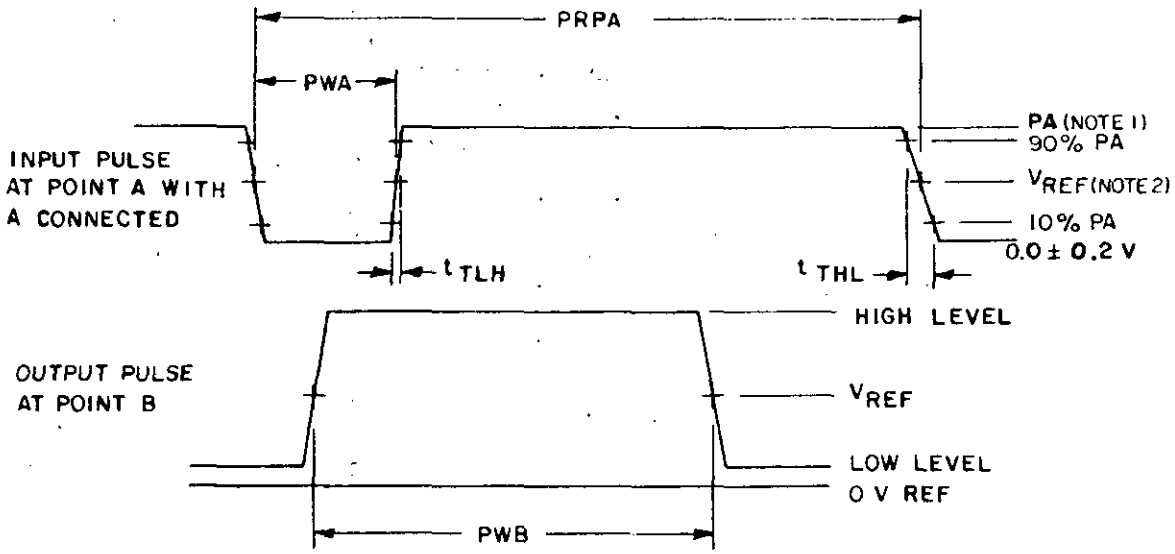
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NOTES:
1. R1 = 51.1Ω ±1%, 1/2 W.
2. R2 = R3 = 68.1 kΩ ±0.1%, 1/8 W.
3. R4 = 3.01 kΩ ±1%, 1/8 W.
4. C1 = C2 = 1,000 pF ±1%.
5. V_{CC} = 5 V ±1% for test conditions P1 and P2.
   = 15 V ±1% for test conditions Q1 and Q2.
6. Set counter to measure time interval(PWB).

FIGURE 18. Initial accuracy of monostable timing test circuit for tables XV and XVI.

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NOTES:
1. Input pulse parameters for item J01:
   PRPA = 90 µs ±1%.
   PWA = 30 µs ±1%.
   PA = 4 V ±1% for test conditions P1 and P2.
   PA = 12 V ±1% for test conditions Q1 and Q2.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. V_{REF} = @ V_{CC} = 5 V (-2.5 ±0.1 V).
   @ V_{CC} = 15 V (-7.5 ±0.1 V).
3. Verify that all outputs have a high level of $\geq$ 3 V for a V_{CC} of 5 V and $\geq$ 13 V for a V_{CC} of 15 V. Also verify that all outputs have a low level of $\leq$ 0.25 V for a V_{CC} of 5 V and $\leq$ 2.25 V for a V_{CC} of 15 V.
4. The physical length of electrical conductors from item Y01 to input pins shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

* FIGURE 19. *Initial accuracy of monostable timing test circuit waveforms for tables XV and XVI.*

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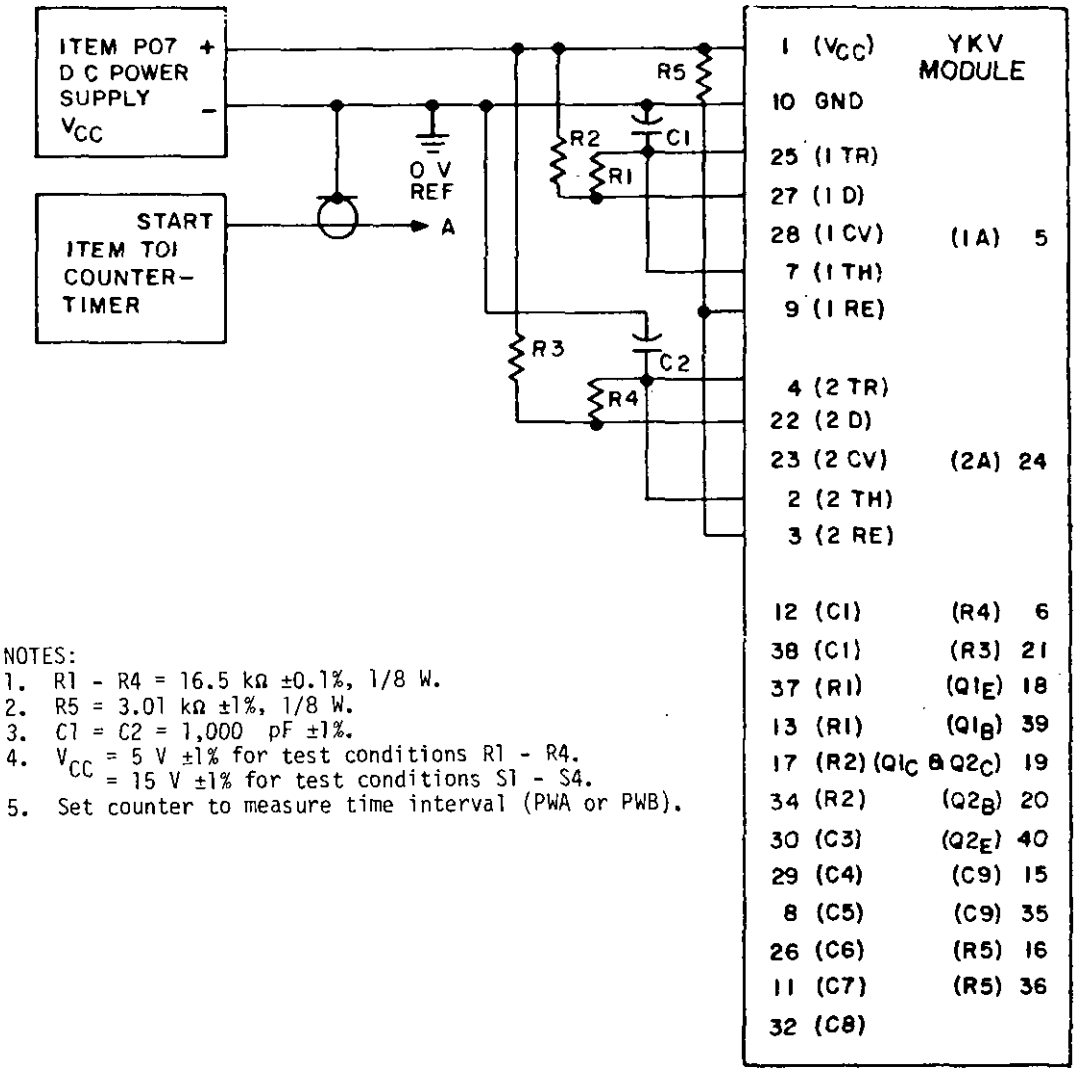
TABLE XVII. Initial accuracy of astable timing test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
R1	OUT 1 (PWA) Initial accuracy of astable timing, V_{CC} = 5 V	5	19.7	24.6	µs	20, 21
R2	OUT 2 (PWA) "	24	19.7	24.6	"	"
R3	OUT 1 (PWB) "	5	9.83	12.3	"	"
R4	OUT 2 (PWB) "	24	9.83	12.3	"	"

TABLE XVIII. Initial accuracy of astable timing test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
S1	OUT 1 (PWA) Initial accuracy of astable timing, V_{CC} = 15 V	5	20.1	25.1	µs	20, 21
S2	OUT 2 (PWA) "	24	20.1	25.1	"	"
S3	OUT 1 (PWB) "	5	10.03	12.50	"	"
S4	OUT 2 (PWB) "	24	10.03	12.50	"	"

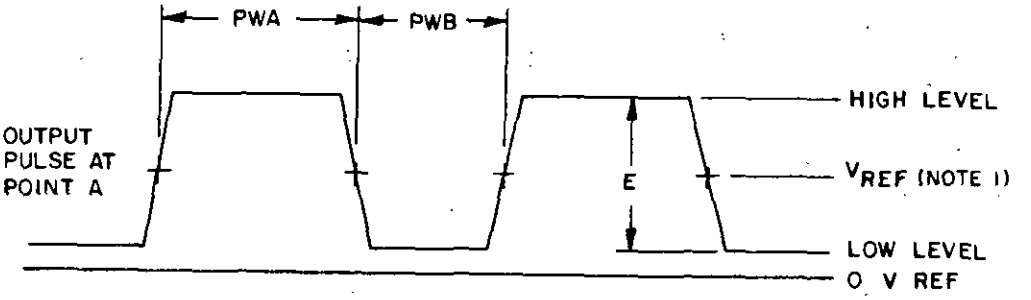
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NOTES:
1. R1 - R4 = 16.5 kΩ ±0.1%, 1/8 W.
2. R5 = 3.01 kΩ ±1%, 1/8 W.
3. C1 = C2 = 1,000 pF ±1%.
4. V_{CC} = 5 V ±1% for test conditions R1 - R4.
   = 15 V ±1% for test conditions S1 - S4.
5. Set counter to measure time interval (PWA or PWB).

FIGURE 20. Initial accuracy of astable timing test circuit for tables XVII and XVIII.

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NOTES:
1. V_{REF} = @ V_{CC} = 5 V (-1.5 ±0.1 V).
   @ V_{CC} = 15 V (-13 ±0.1 V).
2. Verify that all outputs have a high level of $\geq$ 3 V for a V_{CC} of 5 V and $\geq$ 13 V for a V_{CC} of 15 V. Also verify that all outputs have a low level of $\leq$ 0.25 V for a V_{CC} of 5 V and $\leq$ 2.25 V for a V_{CC} of 15 V.
3. Measure PWA for test conditions R1, R2, S1, and S2.
4. Measure PWB for test conditions R3, R4, S3, and S4.
5. All voltages are referenced to ground.

FIGURE 21. Initial accuracy of astable timing test circuit waveform for tables XVII and XVIII.

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TABLE XIX. Monostable output pulse width test.

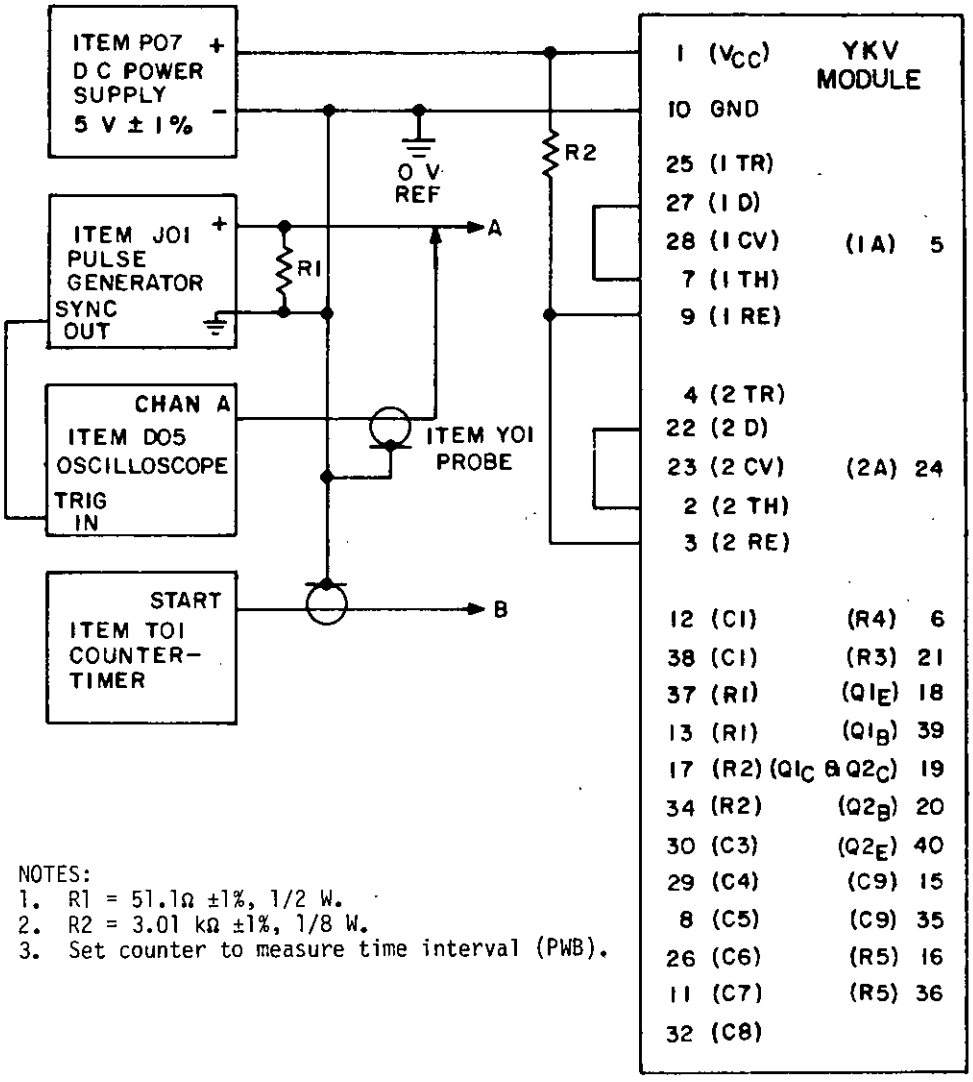
Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 27 to pin	Pin 22 to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
T1	Monostable output pulse width R = 10 kΩ, C = 0.01 μF	25	5	21,26	Open	Open	Open	96.1	124.6	μs	22,23
T2	Monostable output pulse width R = 49.9 kΩ, C = 0.01 μF	"	"	36,26	"	16	"	479.3	622.0	μs	"
T3	Monostable output pulse width R = 100 kΩ, C = 0.01 μF	"	"	6,26	"	Open	"	0.96	1.25	ms	"
T4	Monostable output pulse width R = 10 kΩ, C = 0.1 μF	"	"	21,30	"	Open	"	0.96	1.25	"	"
T5	Monostable output pulse width R = 49.9 kΩ, C = 0.1 μF	"	"	36,30	"	16	"	4.79	6.22	"	"
T6	Monostable output pulse width R = 100 kΩ, C = 0.1 μF	"	"	6,30	"	Open	"	9.6	12.5	"	"
T7	Monostable output pulse width R = 10 kΩ, C = 0.22 μF	"	"	21,29	"	Open	"	2.11	2.74	"	"
T8	Monostable output pulse width R = 49.9 kΩ, C = 0.22 μF	"	"	36,29	"	16	"	10.5	13.7	"	"
T9	Monostable output pulse width R = 100 kΩ, C = 0.22 μF	"	"	6,29	"	Open	"	21.1	27.4	"	"
T10	Monostable output pulse width R = 10 kΩ, C = 0.47 μF	"	"	21,35	"	Open	15	4.51	5.86	"	"
T11	Monostable output pulse width R = 49.9 kΩ, C = 0.47 μF	"	"	36,35	"	16	15	22.5	29.2	"	"
T12	Monostable output pulse width R = 100 kΩ, C = 0.47 μF	"	"	6,35	"	Open	15	45.1	58.6	"	"

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TABLE XIX. Monostable output pulse width test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 27 to pin	Pin 22 to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
T13	Monostable output pulse width R = 10 kΩ, C = 0.01 μF	4	24	Open	21,26	Open	Open	96.1	124.6	μs	22,23
T14	Monostable output pulse width R = 49.9 kΩ, C = 0.01 μF	"	"	"	36,26	16	"	479.3	622.0	μs	"
T15	Monostable output pulse width R = 100 kΩ, C = 0.01 μF	"	"	"	6,26	Open	"	0.96	1.25	ms	"
T16	Monostable output pulse width R = 10 kΩ, C = 0.1 μF	"	"	"	21,30	Open	"	0.96	1.25	"	"
T17	Monostable output pulse width R = 49.9 kΩ, C = 0.1 μF	"	"	"	36,30	16	"	4.79	6.22	"	"
T18	Monostable output pulse width R = 100 kΩ, C = 0.1 μF	"	"	"	6,30	Open	"	9.6	12.5	"	"
T19	Monostable output pulse width R = 10 kΩ, C = 0.22 μF	"	"	"	21,29	Open	"	2.11	2.74	"	"
T20	Monostable output pulse width R = 49.9 kΩ, C = 0.22 μF	"	"	"	36,29	16	"	10.5	13.7	"	"
T21	Monostable output pulse width R = 100 kΩ, C = 0.22 μF	"	"	"	6,29	Open	15	21.1	27.4	"	"
T22	Monostable output pulse width R = 10 kΩ, C = 0.47 F	"	"	"	21,35	Open	"	4.51	5.86	"	"
T23	Monostable output pulse width R = 49.9 kΩ, C = 0.47 F	"	"	"	36,35	16	"	22.5	29.2	"	"
T24	Monostable output pulse width R = 100 kΩ, C = 0.47 F	"	"	"	6,35	Open	"	45.1	58.6	"	"

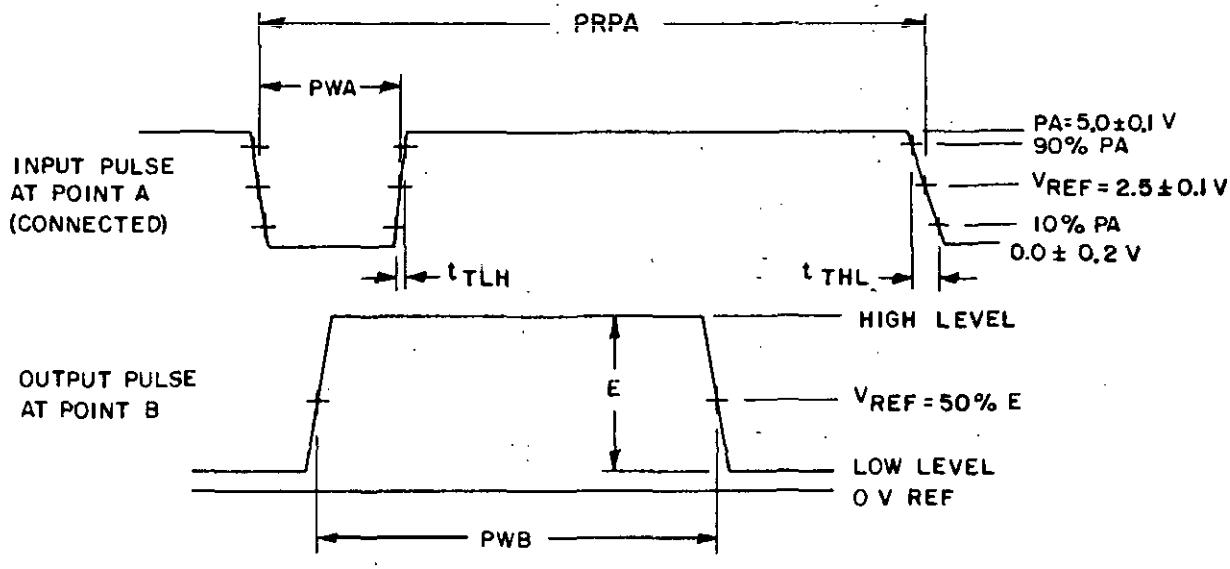
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NOTES:
1. R1 = 51.1Ω ±1%, 1/2 W.
2. R2 = 3.01 kΩ ±1%, 1/8 W.
3. Set counter to measure time interval (PWB).

FIGURE 22. Monostable output pulse width test circuit for table XIX.

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NOTES:
1. Input pulse parameters for item J01:
   PRPA = 65.0 ms ±1%.
   PWA = 75 µs ±1%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify that all inputs have a high level of ≥ 3 V and a low level of ≤ 0.25 V.
3. The physical length of electrical conductors from item Y01 to input pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 23. Monostable output pulse width test circuit waveforms for table XIX.

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* TABLE XX. Astable output pulse characteristics test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 25 to pin	Pin 27 to pin	Min	Max		
U1	OUT 1 (PWA) Astable output pulse characteristics	5	26, 34	21, 17	110.5	170.2	µs	24, 25
U2	OUT 1 (PWB) "	"	26, 34	21, 17	55.3	85.1	"	"
U3	OUT 1 (PWA) "	"	26, 16	21, 36	331.3	510.0	"	"
U4	OUT 1 (PWB) "	"	26, 16	21, 36	275.9	424.8	"	"
U5	OUT 1 (PWA) "	"	26, 37	6, 13	1.11	1.70	ms	"
U6	OUT 1 (PWB) "	"	26, 37	6, 13	0.553	0.851	"	"
U7	OUT 1 (PWA) "	"	30, 34	21, 17	1.11	1.70	"	"
U8	OUT 1 (PWB) "	"	30, 34	21, 17	0.553	0.851	"	"
U9	OUT 1 (PWA) "	"	30, 16	21, 36	3.31	5.10	"	"
U10	OUT 1 (PWB) "	"	30, 16	21, 36	2.76	4.25	"	"
U11	OUT 1 (PWA) "	"	30, 37	6, 13	11.1	17.0	"	"
U12	OUT 1 (PWB) "	"	30, 37	6, 13	5.53	8.51	"	"

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* TABLE XX. Astable output pulse characteristics test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 25 to pin	Pin 27 to pin	Pin 15 to pin	Min	Max		
U13	OUT 1 (PWA) Astable output pulse characteristics	5	29, 34	21, 17	Open	2.43	3.74	ms	24, 25
U14	OUT 1 (PWB) "	"	29, 34	21, 17	"	1.22	1.87	"	"
U15	OUT 1 (PWA) "	"	29, 16	21, 36	"	7.30	11.23	"	"
U16	OUT 1 (PWB) "	"	29, 16	21, 36	"	6.08	9.36	"	"
U17	OUT 1 (PWA) "	"	29, 37	6, 13	"	24.3	37.4	"	"
U18	OUT 1 (PWB) "	"	29, 37	6, 13	"	12.2	18.7	"	"
U19	OUT 1 (PWA) "	"	35, 34	21, 17	10	5.20	8.00	"	"
U20	OUT 1 (PWB) "	"	35, 34	21, 17	"	2.60	4.00	"	"
U21	OUT 1 (PWA) "	"	35, 16	21, 36	"	15.5	23.9	"	"
U22	OUT 1 (PWB) "	"	35, 16	21, 36	"	12.9	20.0	"	"
U23	OUT 1 (PWA) "	"	35, 37	6, 13	"	52.0	80.0	"	"
U24	OUT 1 (PWB) "	"	35, 37	6, 13	"	26.0	40.0	"	"

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* TABLE XX. Astable output pulse characteristics test - Continued.

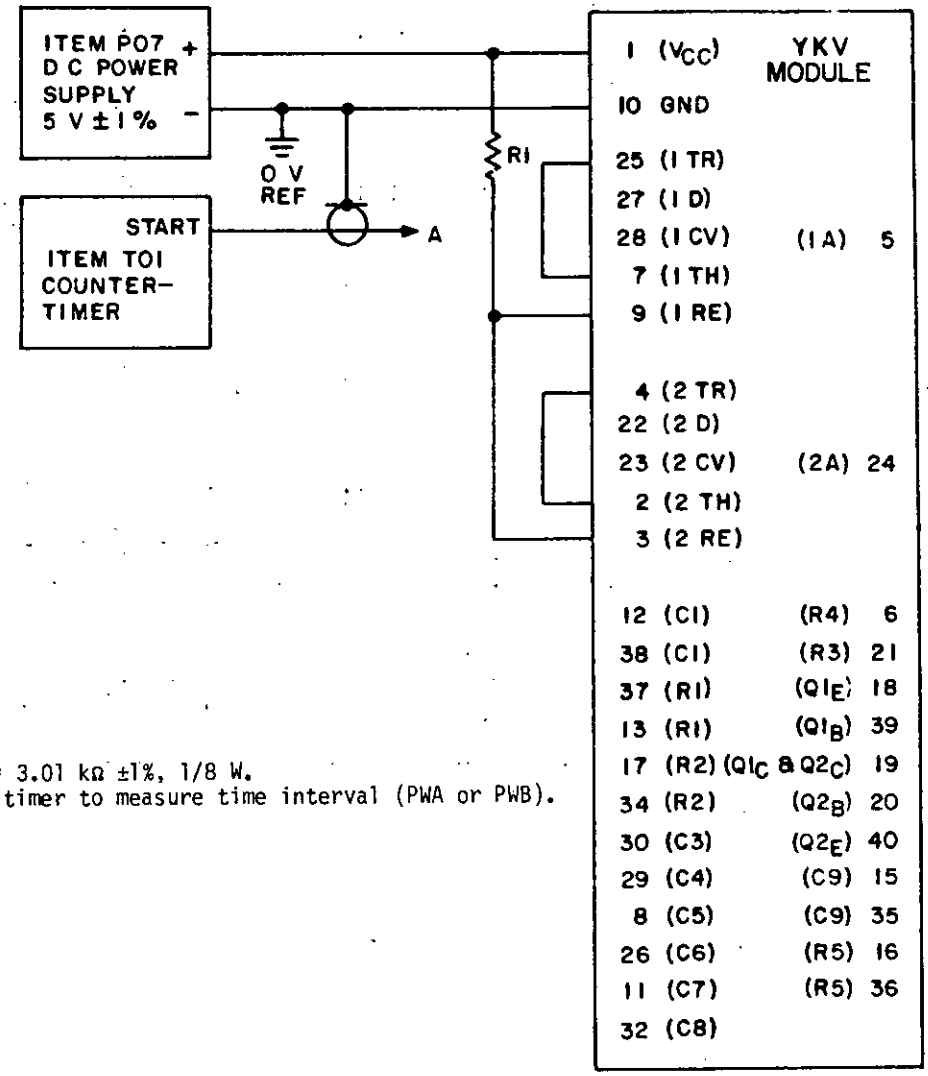
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 25 to pin	Pin 27 to pin	Min	Max		
U25	OUT 2 (PWA) Astable output pulse characteristics	24	26, 34	21, 17	110.5	170.2	µs	24, 25
U26	OUT 2 (PWB) "	"	26, 34	21, 17	55.3	85.1	"	"
U27	OUT 2 (PWA) "	"	26, 16	21, 36	331.3	510.0	"	"
U28	OUT 2 (PWB) "	"	26, 16	21, 36	275.9	424.8	"	"
U29	OUT 2 (PWA) "	"	26, 37	6, 13	1.11	1.70	ms	"
U30	OUT 2 (PWB) "	"	26, 37	6, 13	0.553	0.851	"	"
U31	OUT 2 (PWA) "	"	30, 34	21, 17	1.11	1.70	"	"
U32	OUT 2 (PWB) "	"	30, 34	21, 17	0.553	0.851	"	"
U33	OUT 2 (PWA) "	"	30, 16	21, 36	3.31	5.10	"	"
U34	OUT 2 (PWB) "	"	30, 16	21, 36	2.76	4.25	"	"
U35	OUT 2 (PWA) "	"	30, 37	6, 13	11.1	17.0	"	"
U36	OUT 2 (PWB) "	"	30, 37	6, 13	5.53	8.51	"	"

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* TABLE XX. Astable output pulse characteristics test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 4 to pin	Pin 22 to pin	Pin 15 to pin	Min	Max		
U37	OUT 2 (PWA) Astable output pulse characteristics	24	29, 34	21, 17	Open	2.43	3.74	ms	24,25
U38	OUT 2 (PWB) "	"	29, 34	21, 17	"	1.22	1.87	"	"
U39	OUT 2 (PWA) "	"	29, 16	21, 36	"	7.30	11.23	"	"
U40	OUT 2 (PWB) "	"	29, 16	21, 36	"	6.08	9.36	"	"
U41	OUT 2 (PWA) "	"	29, 37	6, 13	"	24.3	37.4	"	"
U42	OUT 2 (PWB) "	"	29, 37	6, 13	"	12.2	18.7	"	"
U43	OUT 2 (PWA) "	"	35, 34	21, 17	10	5.20	8.00	"	"
U44	OUT 2 (PWB) "	"	35, 34	21, 17	"	2.60	4.00	"	"
U45	OUT 2 (PWA) "	"	35, 16	21, 36	"	15.5	23.9	"	"
U46	OUT 2 (PWB) "	"	35, 16	21, 36	"	12.9	20.0	"	"
U47	OUT 2 (PWA) "	"	35, 37	6, 13	"	52.0	80.0	"	"
U48	OUT 2 (PWB) "	"	35, 37	6, 13	"	26.0	40.0	"	"

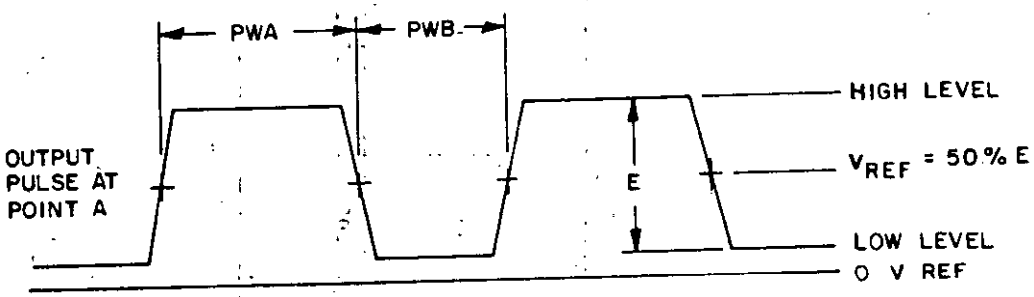
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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. Set timer to measure time interval (PWA or PWB).

FIGURE 24. Astable output pulse characteristic test circuit for table XX.

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NOTES:
1. Verify that all outputs have a output $\geq$ 3 V and a low level of $\leq$ 0.25 V.
2. Measure PWA for test conditions 1 - 47 (odd numbers).
3. Measure PWB for test conditions 2 - 48 (even numbers).
4. All voltages are referenced to ground.

FIGURE 25. Astable output pulse characteristics test circuit waveforms for table XX.

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TABLE XXI. Maximum frequency and minimum trigger pulse check test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 27 to pin	Pin 22 to pin	Min	Max		
V1	OUT 1 Maximum frequency and minimum trigger pulse check	25	5	21, 11	Open	12.9	13.7	µs	26,27
V2	OUT 2 Maximum frequency and minimum trigger pulse check	4	24	Open	21, 11	12.9	13.7	µs	26,27

TABLE XXII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 27 to pin	Pin 22 to pin	Min	Max		
W1	OUT 1 Transition time (t_{TLH}), C_L = 50 pF	25	5	Open	7,21,32	Open	10	300	ns	28, 29
W2	OUT 2 Transition time (t_{TLH}), C_L = 50 pF	4	24	Open	Open	2,21,32	10	300	ns	28, 29

TABLE XXIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 27 to pin	Pin 22 to pin	Min	Max		
X1	OUT 1 Transition time (t_{THL}), C_L = 50 pF	25	5	Open	7,21,32	Open	10	300	ns	28, 29
X2	OUT 2 Transition time (t_{THL}), C_L = 50 pF	4	24	Open	Open	2,21,32	10	300	ns	28, 29

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TABLE XXIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 27 to pin	Pin 22 to pin	Min	Max		
Y1	OUT 1 Propagation delay time (t_{PLH}), C_L = 50 pF (trigger-to-output)	25	5	25	7,21,32	Open	25	800	ns	28,29
Y2	OUT 2 Propagation delay time (t_{PLH}), C_L = 50 pF (trigger-to-output)	4	24	4	Open	2,21,32	25	800	ns	28,29

TABLE XXV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	OUT 1 Propagation delay time (t_{PHL}), C_L = 50 pF (reset-to-output)	5	6, 26	Open	0.2	1.5	µs	31, 32
Z2	OUT 2 Propagation delay time (t_{PHL}), C_L = 50 pF (reset-to-output)	24	Open	6, 26	0.2	1.5	µs	31, 32

TABLE XXVI. Output crosstalk (low-level) test.

Test condition	Parameter	Connect							Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 1 to pin	Pin 27 to pin	Pin 25 to pin	Pin 10 to pin	Pin 22 to pin	Pin 4 to pin	Min	Max		
AA1	OUT 2 Output crosstalk, low level	24	9	21,37	7,13,11	3	Open	Open	0	400	mV	33,34
AA2	OUT 2 Output crosstalk, low level	5	3	Open	Open	9	21, 37	2,13,11	0	400	mV	33,34

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TABLE XXVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect							Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Pin 1 to pin	Pin 27 to pin	Pin 25 to pin	Pin 10 to pin	Pin 22 to pin	Pin 4 to pin	Min	Max		
BB1	OUT 2 Output crosstalk, high level	24	9,3	21,37	7,13,11	4	Open	10	0	400	mV	33,34
BB2	OUT 2 Output crosstalk, high level	5	3,9	Open	10	25	21,37	2,13,11	0	400	mV	33,34

TABLE XXVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
CC	Power supply transient amplitude	0	400	mV	35, 36

TABLE XXIX. Power supply filter capacitance test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
DD	Power supply filter capacitance	Open	Open	1	10	0.7	1.3	µF	37

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TABLE XXX. Input capacitance test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
EE1	1TR Input capacitance	25	10	Open	Open	2	15	pF	37
EE2	1D "	27	"	"	"	"	25	"	"
EE3	1CV "	28	"	"	"	"	15	"	"
EE4	1TH "	7	"	"	"	"	"	"	"
EE5	1RE "	9	"	"	"	"	"	"	"
EE6	2TR "	4	"	"	"	"	"	"	"
EE7	2D "	22	"	"	"	"	25	"	"
EE8	2CV "	23	"	"	"	"	15	"	"
EE9	2TH "	2	"	"	"	"	"	"	"
EE10	2RE "	3	"	"	"	"	"	"	"

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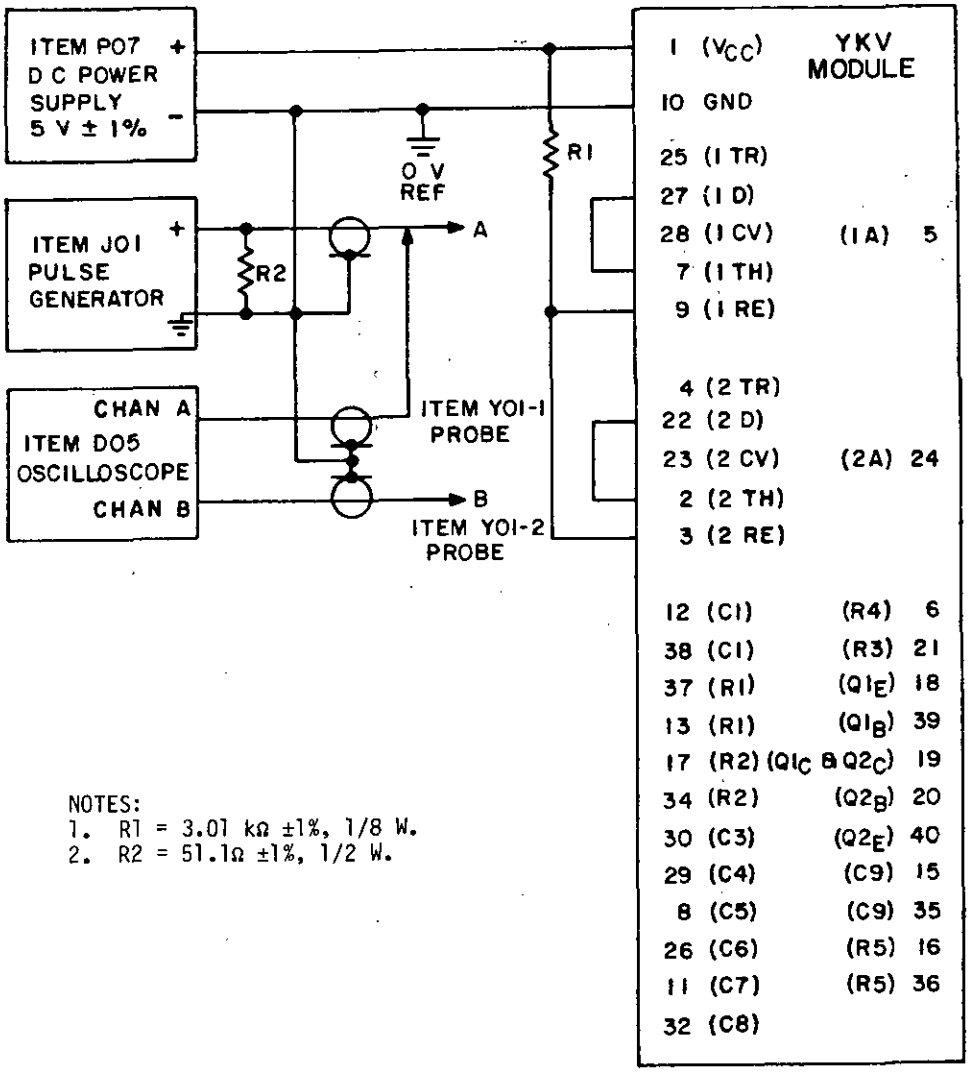
* TABLE XXXI. Capacitance test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	C1	Capacitance	12	38	Open	Open	243	297	pF	37
FF2	C3	"	Open	Open	30	10	0.090	0.110	µF	"
FF3	C4	"	"	"	29	"	0.198	0.242	"	"
FF4	C5	"	"	"	8	"	0.0090	0.0110	"	"
FF5	C6	"	"	"	26	"	0.0090	0.0110	"	"
FF6	C7	"	11	10	Open	Open	90	110	pF	"
FF7	C8	"	Open	Open	32	10	90	110	"	"
FF8	C9	"	Open	Open	15	35	0.423	0.517	"	"

* TABLE XXXII. Resistance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
GG1	R1	Resistance	13	37	98.5	101.5	kΩ	38
GG2	R2	"	34	17	9.85	10.15	"	"
GG3	R3	"	21	1	9.85	10.15	"	"
GG4	R4	"	6	1	98.5	101.5	"	"
GG5	R5	"	16	36	49.15	50.65	"	"

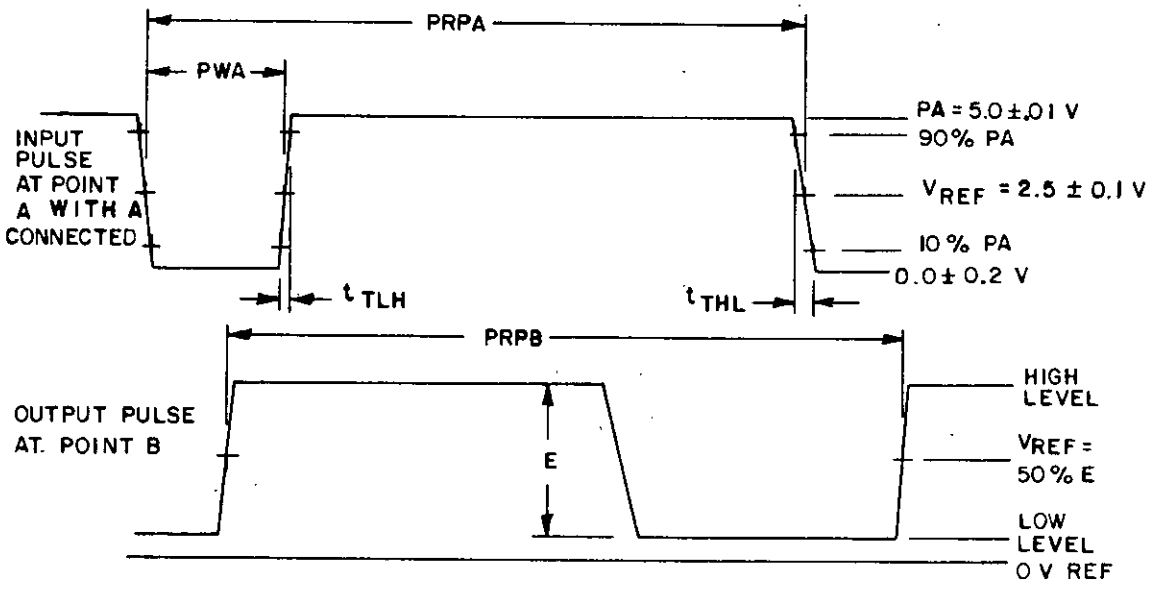
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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. R2 = 51.1Ω ±1%, 1/2 W.

FIGURE 26. Maximum frequency and minimum trigger pulse check test circuit for table XXI.

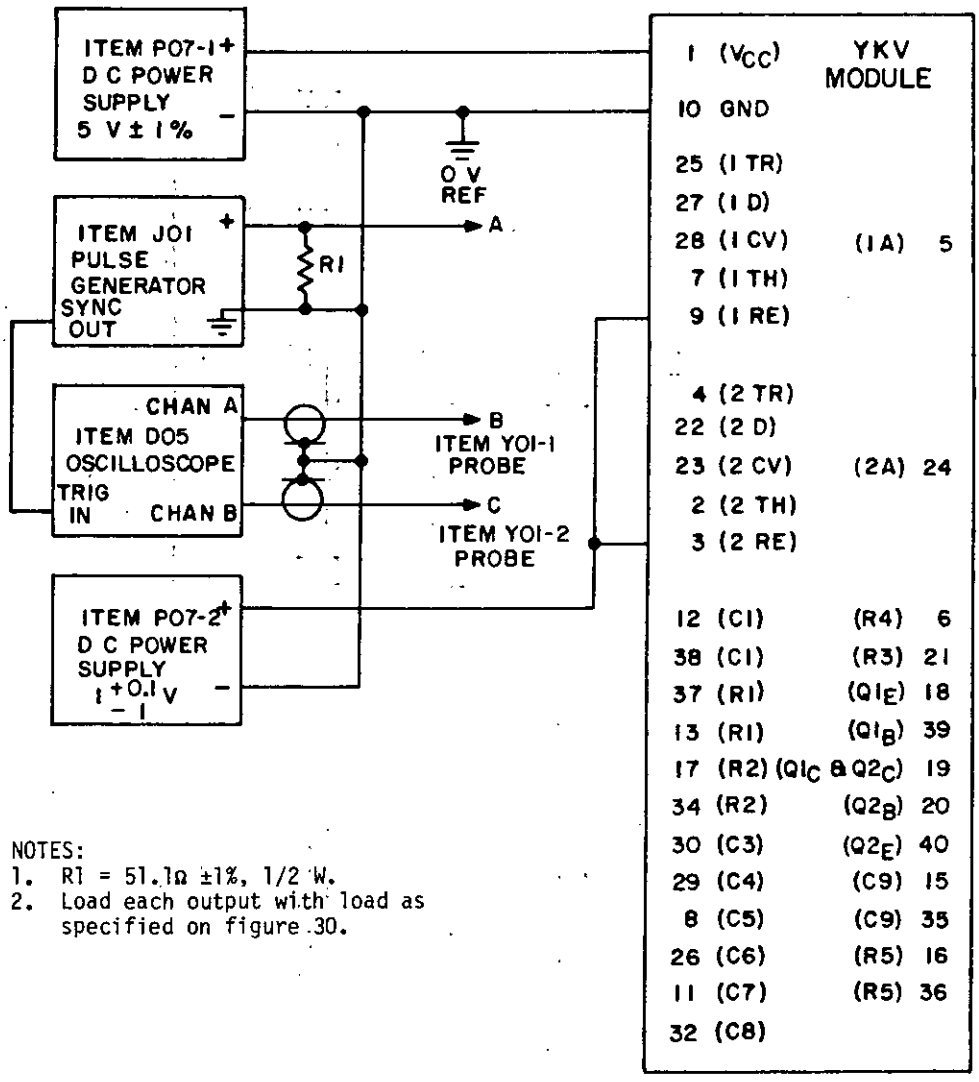
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NOTES:
1. Input pulse parameters for item J01:
   PRPA = 13.3 ±0.2 µs.
   PWA = 200 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify that all inputs have a high level ≥ 3 V and a low level ≤ 0.25 V.
3. The physical length of electrical conductors from items Y01-1 and Y01-2 to input and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

★ FIGURE 27. Maximum frequency and minimum trigger pulse check test circuit waveforms for table XXI.

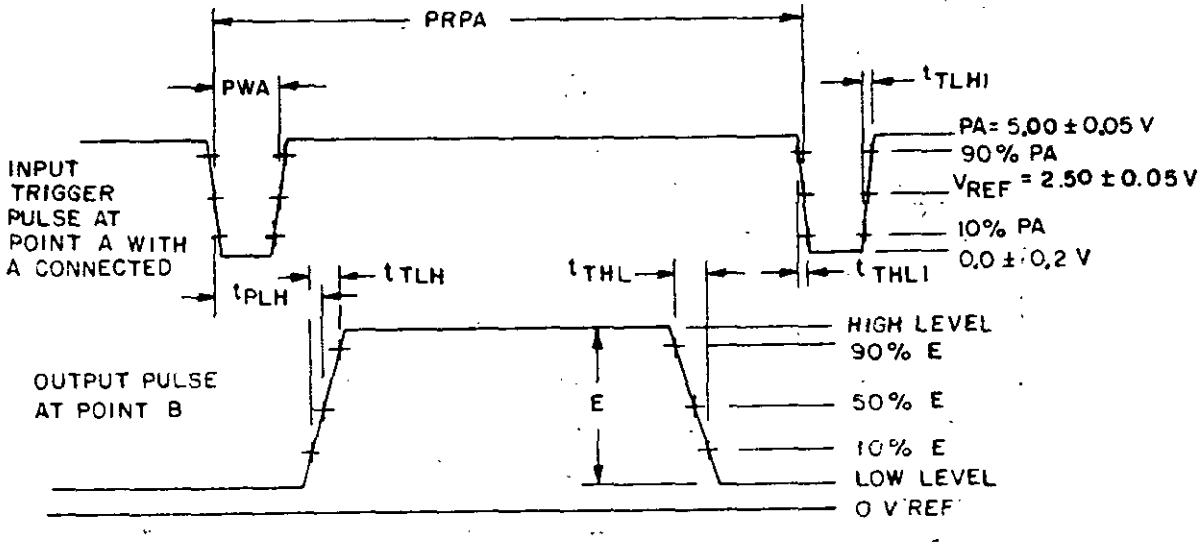
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NOTES:
1. R1 = 51.1Ω ±1%, 1/2 W.
2. Load each output with load as specified on figure 30.

FIGURE 28. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH}) test circuit for tables XXII, XXIII and XXIV.

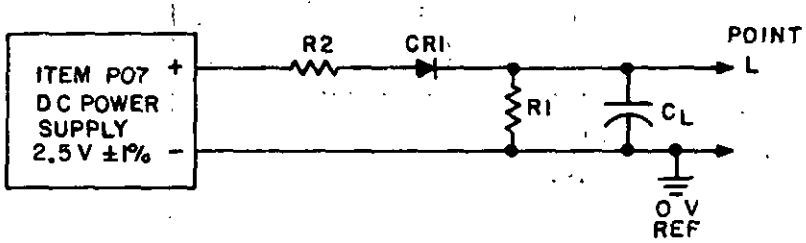
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:19Z
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NOTES:
1. Input pulse parameters for item J01:
   PRPA = 13.3 ±0.2 µs.
   PWA = 400 ±40 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Verify that all outputs have a high level ≥ 3 V and a low level ≤ 0.25 V before making measurement.
3. The physical length of electrical conductors from the resistor R1 to input pins, from items Y01-1 and Y01-2 to input pins, and from load circuits to output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 29. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH}) test circuit waveforms for tables XXII, XXIII, and XXIV.

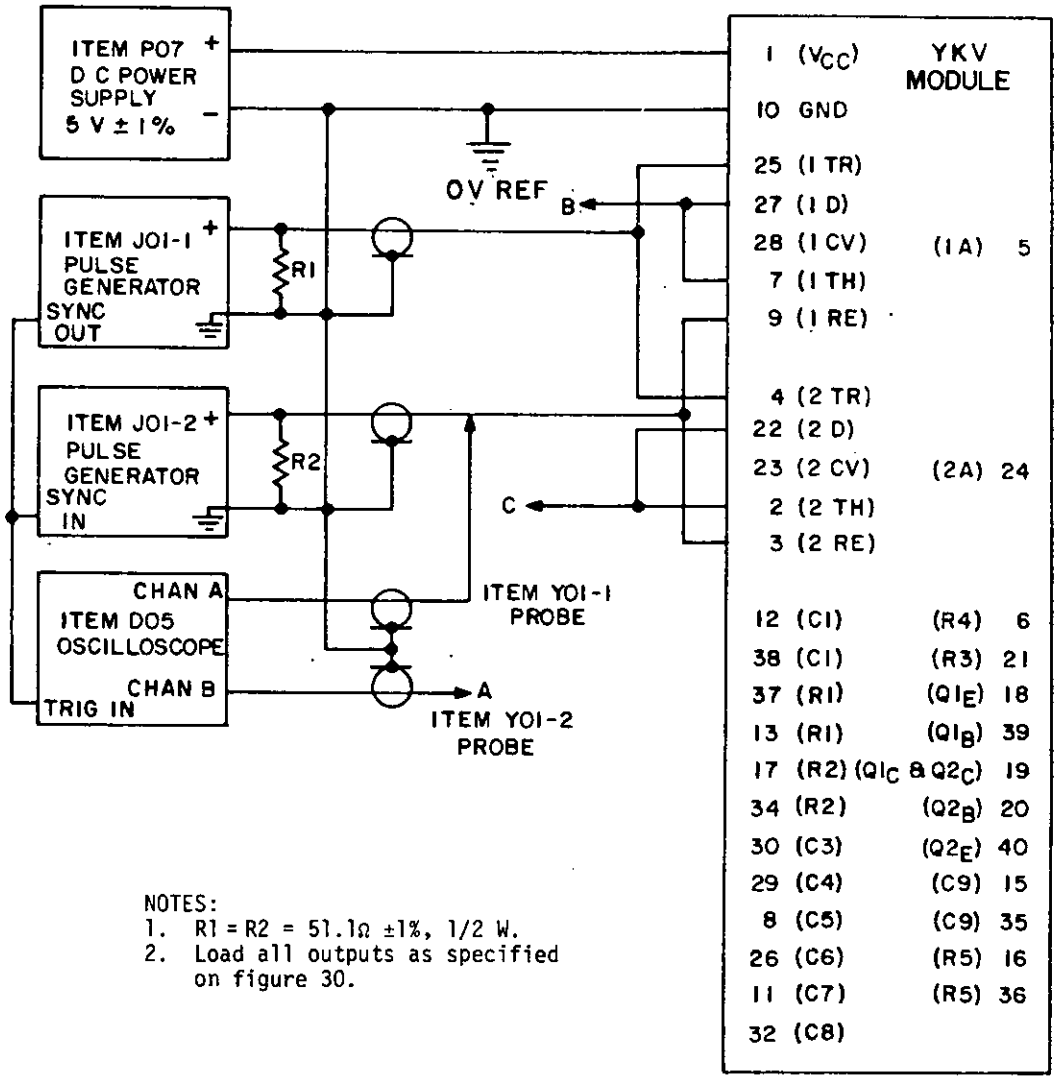
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NOTES:
1. CR1 = 1N4150 or equivalent.
2. R1 = 30.1Ω ±1%, 1/2 W.
3. R2 = Select value for a current flow of 8 mA ±5% out of point L with point L held at 0.25 V.
4. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).
5. C_L = 50 ±5 pF.

FIGURE 30. YKV logic load circuit.

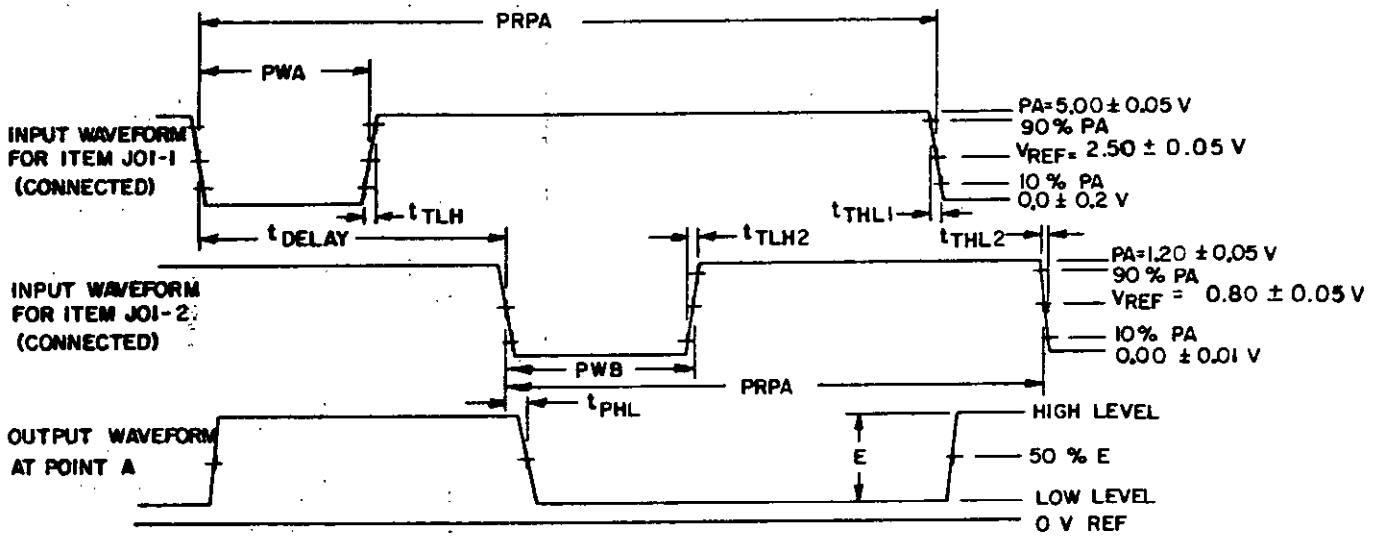
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/2 W.
2. Load all outputs as specified on figure 30.

FIGURE 31. Propagation delay time (t_{PHL}) test circuit for table XXV.

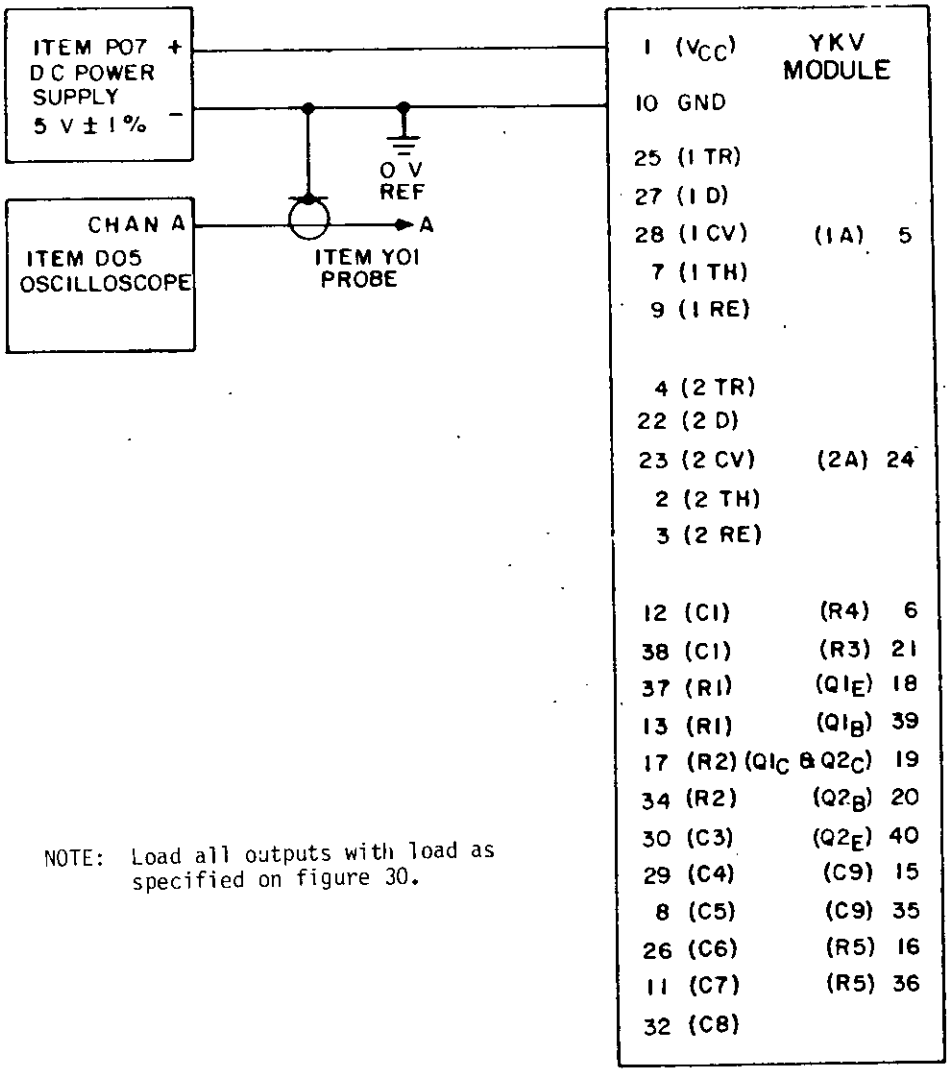
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NOTES:
1. Input pulse parameters for item J01-1:
   PRPA = 14.0 ±1.4 µs.
   PWA = 3 µs ±10%.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
2. $t_{DELAY} = 5$ µs ±1%.
3. Input pulse parameters for item J01-2:
   PRPA = 14.0 ±1.4 µs.
   PWB = 3 µs ±10%.
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.

FIGURE 32. Propagation delay time (t_{PHL}) test circuit waveforms for table XXV.

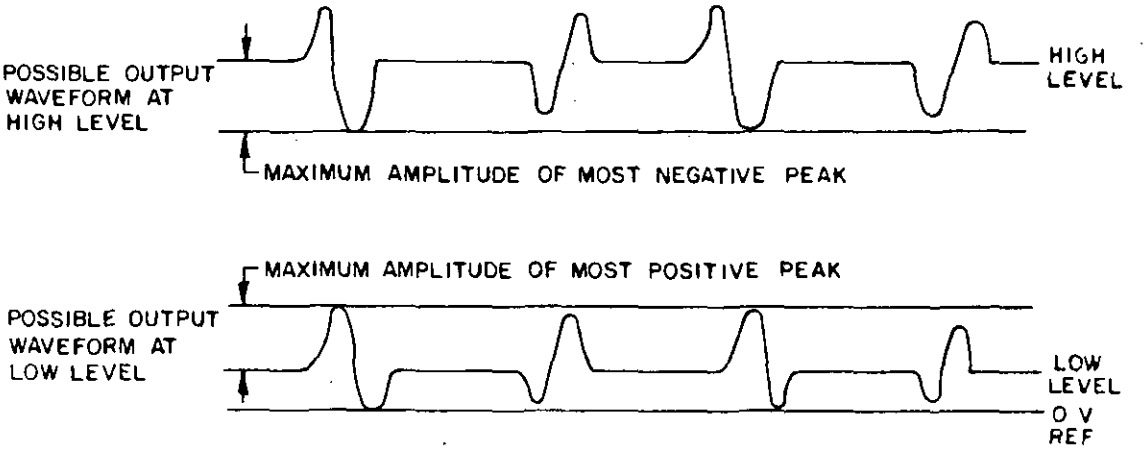
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NOTE: Load all outputs with load as specified on figure 30.

FIGURE 33. Output crosstalk (low-level and high-level) test circuit for tables XXVI and XXVII.

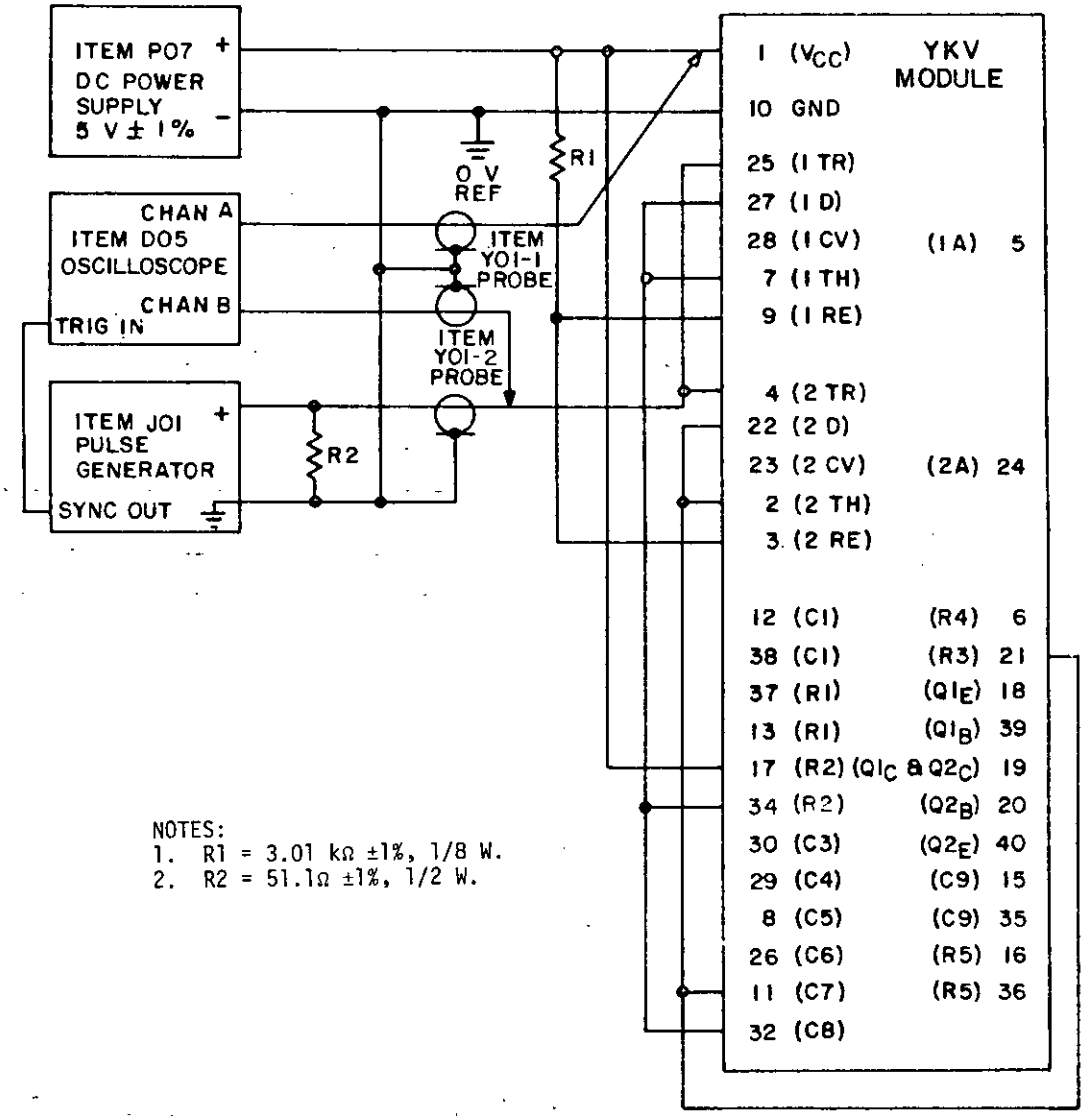
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NOTES:
1. Before making any measurements, verify that the output low level and high level are $\leq$ 0.25 V and $\geq$ 3.2 V respectively.
2. The lead length of electrical conductors between item YO1 and output pins and between load circuit and output pin shall be less than 1 inch (25.4 mm).

FIGURE 34. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXVI and XXVII.

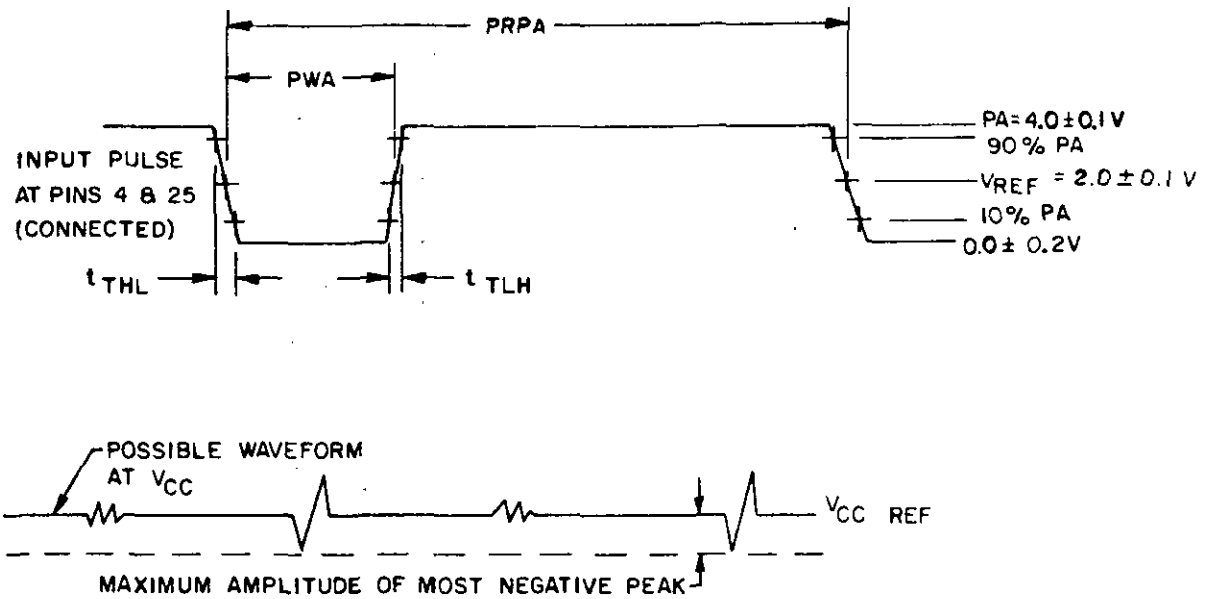
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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. R2 = 51.1Ω ±1%, 1/2 W.

FIGURE 35. Power supply transient amplitude test circuit for table XXVIII.

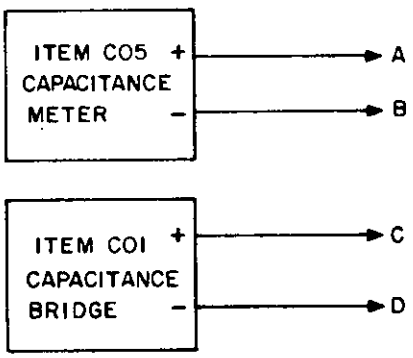
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NOTES:
1. Input pulse parameters for item J01:
   PRPA = 13 µs ±10%.
   PWA = 6.5 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify the input pulse parameters at each input pin before making measurement.
3. Verify the output voltages low levels and high levels are $\leq$ 0.25 V and $\geq$ 3.2 V respectively.
4. All voltages are referenced to ground.

FIGURE 36. Power supply transient amplitude test circuit waveforms for table XVIII.

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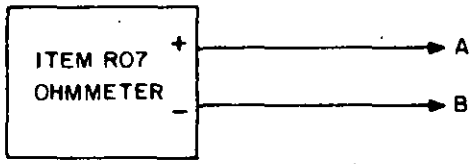


1	(V_{CC})	YKV MODULE	
10	GND		
25	(1 TR)		
27	(1 D)		
28	(1 CV)	(1A)	5
7	(1 TH)		
9	(1 RE)		
4	(2 TR)		
22	(2 D)		
23	(2 CV)	(2A)	24
2	(2 TH)		
3	(2 RE)		
12	(C1)	(R4)	6
38	(C1)	(R3)	21
37	(R1)	($Q1_E$)	18
13	(R1)	($Q1_B$)	39
17	(R2)	($Q1_C$ & $Q2_C$)	19
34	(R2)	($Q2_B$)	20
30	(C3)	($Q2_E$)	40
29	(C4)	(C9)	15
8	(C5)	(C9)	35
26	(C6)	(R5)	16
11	(C7)	(R5)	36
32	(C8)		

NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to the other pins may effect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 37. Power supply filter capacitance, input capacitance, and capacitance test circuit for tables XXIX, XXX, and XXXI.

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YKV MODULE		
1 (V_{CC})		
10 GND		
25 (1 TR)		
27 (1 D)		
28 (1 CV)	(1A)	5
7 (1 TH)		
9 (1 RE)		
4 (2 TR)		
22 (2 D)		
23 (2 CV)	(2A)	24
2 (2 TH)		
3 (2 RE)		
12 (C1)	(R4)	6
38 (C1)	(R3)	21
37 (R1)	($Q1_E$)	18
13 (R1)	($Q1_B$)	39
17 (R2) ($Q1_C$ &	$Q2_C$)	19
34 (R2)	($Q2_B$)	20
30 (C3)	($Q2_E$)	40
29 (C4)	(C9)	15
8 (C5)	(C9)	35
26 (C6)	(R5)	16
11 (C7)	(R5)	36
32 (C8)		

FIGURE 38. Resistance test circuit for table XXXII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/274B(EC)	Modules, Standard Electronic Timer, Digital, Key Code YKV

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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